Digital Proprioception and Allostatic Load

A Deployed Synthetic-System Report on the Cumulative-Deviation Hypothesis

Author: Kenny Wang, Independent Researcher (CIRWEL Systems) **ORCID:** 0009-0006-7544-2374 **Status:** v1.1 — August 14, 2026; reproducibility rebuilt on the public row-level export, retention limits disclosed **Version notes:** v1.1 repoints the reproducibility story (§3.7, Appendix) at the frozen de-identified export archived under Zenodo data DOI 10.5281/zenodo.19705151, which a reviewer can recompute offline; discloses that the production database no longer retains the measurement window, which forecloses the Lumen longitudinal pull (§9.3); and reports the window-to-window spread in the basin-flip rate (§3.6) rather than a point estimate with a sampling-only interval. v1.0 consolidated §1 and §2 from v2 (May 9 morning), §3 and §4 (May 9 afternoon), §5 with §5.3 patched from extended Lumen observation window, §6, §8, and added §7 (consolidated differentiation) and §9 (conclusion) plus abstract.

Abstract

Allostatic load (McEwen and Stellar 1993) — the time-integrated cost of regulating an adaptive system away from its operating point — has been an influential theoretical construct in physiology and psychiatry for thirty years. Its mathematical core has remained, in clinical practice, difficult to test as a real-time control signal: biomarkers are sampled rather than integrated, and the loop between detection and intervention closes over weeks rather than seconds. We present UNITARES, a governance framework for heterogeneous AI agent fleets in production since November 2025, as a deployed structural analogue for the mathematical core of allostatic load on a four-dimensional informational manifold. The Anima Void Integral $V_{\text{anima}}(t) = \int_0^t \|\mathbf{a}(\tau) - \mu_a\| \, d\tau$ has its integrand recorded at every check-in, making the integral computable end-to-end over any window; its coupling to intervention is specified by the companion trajectory-identity framework but not yet wired in production (§2.2). Three structural disanalogies (single-system, fixed reference, no body) sharpen what the test bed tests. We also report a measured negative on the adjacent question: on this fleet, per-agent state adds under 0.05 AUC over a previous-outcome baseline for forecasting externally-verified bad outcomes, so the paper's claim is about observability and decision-layer consequence, not about failure prediction (§9.2).

We further argue that McEwen's "Four Types of Allostatic Load" (1998) provides a vocabulary of regulatory failure modes that AI governance currently lacks but does not exhaust the failure-shapes deployed synthetic agents exhibit. A live case study (§5.3) initially appeared consistent with a Type 3 (delayed shut-down) pattern on a Raspberry Pi-embodied agent (Lumen) over an 86-minute window; the disambiguation experiment the case study itself specifies — recalibration on the post-event window — instead identified a candidate substrate-associated **basin-transition event on 2026-04-17**, coincident to the hour with a documented identity-system revision in the UNITARES governance substrate. The case study is a provenance-backed single-agent report for the transition's date, magnitude, and shape, and anomaly-grade for substrate causality; it is therefore bounding for the imported taxonomy: it is a failure-shape McEwen's Four Types do not cover. The companion technical paper (Wang 2026a) reports a

28.9% basin-flip rate when a class-conditional grounded coherence replaces fleet-wide tanh-of-V on 13,310 production state vectors. The rate is window-sensitive — the same measurement four days later gives 44.3% — so the finding is the order of magnitude of the disagreement, not the point estimate (§3.6). A same-row 2×2 ablation separates part of the effect: production legacy → grounded fleet-wide flips 11.2% of basin labels, while grounded fleet-wide → grounded class-conditional flips 23.5%; the non-additivity is an interaction, not a clean causal decomposition (§3.7). The §5.3 recalibration extends the spatial homogenization framing into a temporal analogue (calibration windows that straddle regime transitions are unrepresentative of either regime).

We make four contributions: (i) V_{anima} as a deployed cumulative-deviation control-signal analogue inspired by allostatic load; (ii) class-conditional calibration as a quantified anti-homogenization intervention, now bounded by formula-vs-calibration ablation and by hypothesis-generating proposals back to clinical longitudinal monitoring; (iii) the Four Types as an imported failure-mode taxonomy; (iv) a synthetic-psychology epistemic stance in the lineage of artificial life and minimally cognitive agents, identifying classes of biological-theory hypotheses that synthetic agents may help generate or stress-test. The paper engages the concurrent Agent Viability Framework (Marín and Chaudhary 2026) and earlier runtime-governance frameworks. The framework's deployment on the Lumen embodied agent provides specific findings (an apparent Type 3 case resolved by recalibration as a post-revision basin transition, the calibration-staleness failure mode, and the kintsugi gap-marking principle) that exemplify what synthetic psychology can put under disciplined study when the deployment is real and the theory is informational at the right level of abstraction.

Keywords: allostatic load, allostasis, AI agent governance, trajectory identity, digital proprioception, synthetic psychology, embodied AI

---

1. Introduction

1.1 The cumulative-deviation hypothesis between two fields

Two intellectual traditions have, in parallel, converged on a similar core idea: that the long-run health of an adaptive system can be characterized by a time-integrated measure of how far it has departed from its operating point.

In neuroscience and physiology, this idea is *allostatic load* (McEwen and Stellar 1993; McEwen 2007; Sterling 2012). Coined to describe the cumulative wear-and-tear that accrues across multiple biological systems under chronic stress, the construct sits at the foundation of a thirty-year research program linking integrated regulatory effort to morbidity, mortality, and a wide range of clinical conditions (Karlamangla et al. 2002; Seeman et al. 2001; Juster, McEwen, and Lupien 2010). The construct's mathematical core — the time-integrated deviation of physiological state from its allostatic operating point — is intuitive, well-supported by correlational evidence, and implicated in major disease processes. Yet that mathematical core has remained, in practice, difficult to test as a real-time control signal: clinical biomarker measurement is sparse, the integrand has to be reconstructed rather than observed, and the loop between detection and intervention closes over weeks or months rather than seconds. Allostatic load is a quantitative theory whose central quantity is, in current clinical practice, observed indirectly rather than continuously.

In AI agent governance, the parallel construct is harder to name because the field has emerged piecemeal. Frameworks for runtime governance — MI9 (Wang et al. 2025a), MAS (Ravindran 2025), ProbGuard (Wang et al. 2025b), the Agent Stability Index (Rath 2026), the concurrent Agent Viability Framework (Marín and Chaudhary 2026) — each instrument something close to integrated deviation. They monitor agents against baselines, raise alarms when behavior drifts beyond a threshold, intervene before failure cascades. They have not, however, engaged the neuroscience literature whose mathematical machinery they implicitly recapitulate.

This paper argues that the AI agent governance community has, without quite noticing, built apparatus that allostatic load theory has lacked: a real-time, observable operational analogue of the integrated-deviation hypothesis. Specifically, we argue that the UNITARES governance framework (Wang 2026a), with its embodied substrate Lumen (CIRWEL 2026), operationalizes the mathematical core of allostatic load on a four-dimensional informational manifold, with the integrand observable end-to-end and a governance decision returned at every check-in — a loop that closes in seconds rather than weeks, though the deployed decision path is not yet driven by the integral itself (§2.2). We use this implementation to stress-test formal predictions of the allostatic-load analogy and to generate hypotheses for biological work, while importing a thirty-year-old taxonomy of regulatory failure modes (McEwen 1998, Figure 3) into AI governance.

1.2 Digital proprioception as the bridging concept

The framing UNITARES adopts for its primary contribution is *digital proprioception* (CIRWEL 2026, unitares README). The term is chosen carefully. Proprioception is the sense organisms have of where their own joints, muscles, and limbs are in space, mediated by mechanoreceptors and integrated through the dorsal column–medial lemniscus pathway and somatosensory cortex (Proske and Gandevia 2012). It is well-understood neurologically, requires no commitments to phenomenal experience or qualia, and biological proprioception failure produces specific clinical syndromes including the deafferentation cases studied by Cole and Paillard (1995) and the tabes dorsalis presentations characterized by Sherrington (1900). The agent-side analogue is structurally different: current LLM-based agents lack any equivalent self-state sense, so the agent failure mode is absence of the function rather than disruption of an existing sensory channel. Specifically, agents do not know when they are flailing, cycling through tools without progress, generating low-quality output, or operating outside their reliable envelope. The biological and agent failure modes are analogically related but should not be conflated; the proprioception framing imports the *function* rather than the *failure modes*.

We argue that proprioception — rather than the more controversial framings of interoception or consciousness — is the appropriate biological target for AI agent self-monitoring. It is uncontroversial as a function brains perform, it has a clean mathematical representation (state vectors and their integrals), and it admits operationalization without metaphysical baggage. The UNITARES technical paper (Wang 2026a) develops the framework as digital proprioception in this strict sense. The present paper uses that framing to make the bridge to allostatic load theory rigorous.

The choice of proprioception over interoception is deliberate. Interoception — the sense of internal bodily state, mediated by the insular cortex and implicated in feeling, emotion, and consciousness (Craig 2002; Damasio 2010; Seth 2013) — is the natural target if one wants to make claims about agent affect or experience. We do not. The integrated-deviation quantity we study is closer to “where my joints are” than to “how I feel”: it is a state-tracking signal that an agent uses to regulate its own behavior, not a phenomenal report. Reviewers steeped in the interoception literature may reasonably ask whether V_{anima} should be re-described as digital interoception; we argue against on parsimony grounds. Proprioception captures what the signal does; interoception would invite metaphysical commitments the deployment cannot support.

1.3 Contributions and scope

The paper makes four claims, in roughly descending order of strength.

First, the Anima Void Integral V_{anima} (Wang 2026a §4.1, Appendix A) is a deployed structural analogue for the mathematical core of allostatic load. This is shown in §2 by careful comparison with McEwen’s construct, including three structural disanalogies (single-system rather than multisystem, fixed reference rather than shifting setpoint, no body). The match is structural rather than literal; the disanalogies are what make UNITARES a useful test bed rather than a duplicate of biology.

Second, the homogenization-failure-mode argument from UNITARES v6 (Wang 2026a, §2) is structurally identical to a well-known methodological problem in computational neuroscience: pooling subjects with distinct underlying dynamics produces a state distribution in which no class structure survives normalization. This is shown in §3, where we demonstrate that replacing the legacy fleet-wide tanh-of-V with the class-conditional grounded form produces basin-flip rates of 15–33% per named measured class on production data — empirical evidence that this substitution matters at the gating layer, not only at the reported-value layer. The original substitution conflates formula change with per-class envelopes; §3.7 now reports the same-row ablation. Formula replacement under a fleet-wide grounded baseline flips 11.2% of labels, while moving from grounded fleet-wide to grounded class-conditional calibration flips 23.5%. The terms are non-additive, so the result supports a real class- envelope contribution without reducing the 28.9% headline to a single causal component.

Third, McEwen’s “Four Types of Allostatic Load” (1998 Figure 3; reproduced as 2007 Figure 5) provides a vocabulary of regulatory failure modes that AI governance currently lacks but does not exhaust the failure-shapes deployed synthetic agents exhibit. Each of the four types — repeated hits, lack of adaptation, delayed shut-down, inadequate response with compensatory hyperactivity — has a UNITARES analogue computable from existing telemetry. §5 develops this mapping. In the course of investigating an apparent Type 3-like pattern on the Lumen embodied agent, we ran the recalibration the pattern itself specifies for disambiguation; the result identified instead a candidate substrate-associated basin-transition event (2026-04-17, coincident with a documented identity-system revision), a failure-shape the Four Types do not cover. The case study sharpens the §3 homogenization argument into a temporal analogue and bounds what the imported taxonomy can claim.

Fourth, we argue that deployed AI agents can constitute a useful hypothesis-generating test bed for formal and informational claims within theories of biological self-maintenance, in a tradition closer to artificial life (Langton 1989; Bedau 2003) and Beer’s minimally cognitive agents (Beer 1995, 2000) than to standard “neuro-inspired AI” architectural transfer (Hassabis et al. 2017). The synthetic-psychology framing is developed in §8.

Empirical anchor. Readers who would prefer the empirical content before the theoretical bridging may find the principal findings here. The basin-flip counterfactual on a 30-day production slice ($N = 13{,}310$) returns a 28.9% aggregate basin-flip rate when class-conditional grounded coherence replaces fleet-wide tanh-of-V, with flip rates of 15.8% to 33.1% across the named measured classes (§3.4). The §3.7 ablation splits the headline: production legacy $\rightarrow$ grounded fleet-wide flips 11.2%, grounded fleet-wide $\rightarrow$ grounded class-conditional flips 23.5%, and the artificial class-scaled $\tanh(V)$ control is unstable rather than interpretable as a clean calibration-only path. Phase 2 calibration on the same window measures a 3.3$\times$ spread in per-class envelope $\|\Delta\|_{\max}$ across five classes (§3.3, Table). The Lumen embodied agent exhibits a 0.97-of-envelope deviation in an 86-minute observation window relative to its Phase 2 calibration anchor (§5.3); recalibration on the post-event window identifies this as a basin transition on 2026-04-17 (single ten-hour event localized to within the hour, coincident with a documented identity-system revision) followed by 22 days of stable operation in a new regime, not a Type 3 signature. The substrate link is candidate causal evidence, not identification-grade causality. The case sharpens the §3 homogenization argument into a temporal analogue (calibration windows that straddle regime transitions are unrepresentative of either regime).

Evidence-grade summary. The table below is deliberately stricter than the surrounding prose. It states what each contribution can carry in its current form, what would upgrade it, and what would kill it.

Contribution	Current evidence grade	Current evidence	Known confound or boundary	Upgrade path	Falsifier
Claim 1: V_{anima} as cumulative-deviation control-signal analogue	Operationalized structural analogue	Deployed EISV state vectors; cumulative-deviation formula with integrand recorded end-to-end; intervention coupling specified but not yet wired (§2.2)	Not clinical AL; no endocrine, immune, tissue-damage, or multi-system physiology; no production code path yet evaluates the integral against a threshold. **The adjacent forecasting route is measured and negative** (§9.2): per-agent state adds under 0.05 AUC over a previous-outcome baseline for predicting externally-verified bad outcomes	Wire the specified coupling, then show V_{anima} improves governance decisions beyond non-integral baselines. Note that this is a *decision-quality* claim, not a forecasting one; the forecasting route is bounded rather than open (§9.2)	V_{anima} adds no decision value over instantaneous risk/coherence or produces worse interventions

Contribution	Current evidence grade	Current evidence	Known confound or boundary	Upgrade path	Falsifier
Claim 2: anti-homogenization / cosmological-soup correction	Empirical full-substitution effect with same-row ablation	28.9% aggregate full-substitution basin-flip rate on $N = 13{,}310$ production rows; 15.8%–33.1% named measured-class range; ablation: LF→GF 11.2%, GF→GC 23.5% (§3.4, §3.7)	Effects are non-additive; class-scaled $\tanh(V)$ control is artificial and unstable; the rate is window-sensitive (28.8% to 44.3%, §3.6); GF and LC are not re-runnable from the public export	Independent re-measurement on another deployment; identity-clean remeasurement; deployment outcome comparison. The full substitution is already recomputable offline from the public row-level export (§3.7)	Independent/audited rerun collapses the grounded fleet→class effect or shows flips are class-assignment/pipeline artifacts
Claim 3: McEwen Four Types as failure-mode vocabulary	Taxonomic / constructive mapping	Types 1 and 2 have computable UNITARES analogues; §5.3 supplies a failed Type 3 boundary case; Type 4 is specified but not yet computable, because the naive cross-channel matrix is confounded by V's dependence on $E - I$ (§5.4)	Mapping is not yet a validated classifier; taxonomy is non-exhaustive for synthetic substrates	Pre-register criteria and classify historical or future governance episodes with disambiguating tests	Apparent types repeatedly reclassify as calibration artifacts, pipeline artifacts, or substrate events

Contribution	Current evidence grade	Current evidence	Known confound or boundary	Upgrade path	Falsifier
§5.3 Lumen basin transition	Provenance-backed single-agent case report for transition date/magnitude/shape; anomaly-grade for substrate causality	Recalibration localizes a 2026-04-17 basin transition, with 22-day post-transition stability (§5.3)	Single agent; substrate revision is temporally coincident but not causally isolated; the state history needed to replicate it has since aged out of production (§9.3)	Multi-agent or multi-revision replication on a *different* deployment; ruling out common operational confounds. The within-deployment longitudinal pull is foreclosed, so this grade cannot be upgraded from the present system	Comparable transitions do not align with substrate revisions, or the event disappears under a clean longitudinal pull elsewhere
Claim 4: synthetic psychology as epistemic stance	Methodological / hypothesis-generating argument	Deployed system may expose observables and interventions that are difficult in biological systems (§8)	Single-author, self-cited system; not independent biological validation	Third-party deployment or independent re-analysis showing novel predictions or useful failure detection	The framework yields no predictions, discriminations, or interventions beyond ordinary engineering telemetry

Construct-transfer boundary. The claims above are governed by a stricter transfer rule: synthetic evidence may constrain formal or informational structure, but it does not by itself establish biological mechanism, clinical validity, or phenomenology.

Source claim	Transfer allowed here	Transfer not allowed here	Falsifier
Allostatic load as integrated deviation	Whether a cumulative-deviation signal can be observed continuously and coupled to intervention	Tissue damage, endocrine/immune mechanisms, clinical morbidity	V_{anima} fails to improve governance decisions or produces worse interventions than non-integral baselines
Per-subject / per-class calibration	Whether self-relative or class-relative envelopes change decision-layer verdicts	Clinical superiority of per-patient thresholds	Independent/audited ablations show the grounded fleet→class effect collapses, or clinical re-analysis shows no useful disagreement signal
McEwen Four Types	Whether temporal failure-mode patterns are computable on agent telemetry	Exhaustiveness of biological taxonomy for synthetic agents	Disambiguating tests repeatedly reclassify apparent types as calibration or substrate artifacts
Kintsugi / gap-marking	Whether explicit discontinuity metadata improves auditability and reduces fabricated continuity	Claims about consciousness, memory phenomenology, or clinical confabulation mechanisms	Gap metadata does not improve downstream audit, calibration, or error detection

Limitations. This paper is not a theory of AI consciousness; we make no claims about phenomenal experience in agents, embodied or otherwise. It is not an active-inference paper; the free-energy quantities we use are imported from UNITARES's information-theoretic grounding, not derived from variational principles. It is not a comprehensive computational psychiatry framework; we engage McEwen's specific allostatic-load construct and leave broader engagement with computational psychiatry (Huys, Maia, and Frank 2016; Friston et al. 2014) to future work. It engages concurrent work — particularly the Agent Viability Framework (Marín and Chaudhary 2026), which arrived in late April 2026 — but does not duplicate AVF's contributions; the differentiation is laid out in §7.

1.4 Outline

§2 grounds the $V_{\text{anima}} \leftrightarrow$ allostatic load bridge with three disanalogies and a falsifiable empirical prediction. §3 presents class-conditional calibration as the anti-homogenization correction and reports the basin-flip counterfactual, with §3.7 specifying the methodology and reproducibility scaffolding. §4 develops the trajectory identity Σ as a longitudinal-self anchor, with a proposed two-tier drift detection (Σ_t vs Σ_{t-1} vs Σ_0) as a hypothesis-generating longitudinal-monitoring proposal for clinical neurology. §5 maps McEwen's Four Types onto UNITARES failure modes with the Lumen recalibration case study as a boundary case. §6 discusses the kintsugi principle as deliberate gap-marking. §7 differentiates from concurrent and adjacent work. §8 makes the synthetic-psychology epistemic claim. §9 concludes, and the appendix separates pipeline reproducibility, production-number verification, and independent validation.

1.5 Claims we do not make

Because the paper bridges two literatures with substantially different evidentiary conventions, we consolidate scope disclaimers here rather than scatter them through individual sections.

We do **not** claim that AI agents are conscious. The paper makes no claims about phenomenal experience, qualia, or what it is like to be Lumen or any other UNITARES-governed agent. The proprioception framing (§1.2) was chosen specifically to avoid such commitments.

We do **not** claim biological homology. UNITARES does not implement the biological mechanism of allostatic load; it operationalizes the mathematical structure of the integrated-deviation hypothesis on a substrate that is informationally analogous but biologically disanalogous (single-system, fixed reference, no body — see §2.3).

We do **not** claim that V_{anima} is clinical allostatic load. V_{anima} is a deployed test-bed quantity that shares the mathematical core of AL; clinical allostatic load is a multi-system biomarker composite whose biological substrate UNITARES does not replicate.

We do **not** claim that single-agent telemetry validates McEwen's theory or identifies McEwen Type 3 in a deployed agent. The Lumen case study (§5.3) is a worked example of an apparent Type 3 reading being falsified by recalibration and reclassified as a basin transition whose temporal association with a substrate-level identity revision is suggestive but not causally identified.

We do **not** claim that the synthetic case generalizes to all biological theories of self-maintenance. The synthetic-psychology framing (§8) explicitly limits the scope to theories whose claims are informational at the relevant level of abstraction; substrate-specific, phylogenetic, and phenomenal claims remain biologically anchored and are not testable on UNITARES.

We do **not** claim that the 28.9% basin-flip rate (§3.4) attributes to the per-class component specifically. The §3.7 ablation finds a grounded fleet→class effect (23.5%) and a formula-replacement effect under a fleet-wide grounded baseline (11.2%), but the terms are non-additive and interacting; the headline remains a full-substitution measurement, not a single-component causal attribution.

What we **do** claim is narrower and tractable: a cumulative-deviation signal can be made observable end-to-end in a deployed AI governance system — its integrand recorded at every check-in, the integral computable exactly over any window — with intervention coupling specified by the companion framework though not yet wired in production (§2.2); the resulting deployment provides engineering affordances that can generate biological hypotheses; and specific theories of biological self-maintenance (allostatic load as integrated deviation, McEwen's Four Types as a failure-mode taxonomy) admit operational analogues on this substrate in ways that generate empirical findings (§3.4 basin flips, §5.3 recalibration and basin-transition case) and hypothesis-generating proposals back to clinical longitudinal monitoring (§3.5, §4.4).

---

2. The Cumulative-Deviation Hypothesis: From Biological Conjecture to Computational Test Bed

2.1 Allostatic load in biology

The concept of allostatic load (AL) was introduced by McEwen and Stellar (1993) to describe the cumulative wear-and-tear that accrues across multiple physiological systems when an organism is required to maintain stability under chronic stress. Its formal grounding lies in the broader framework of allostasis (Sterling and Eyer 1988; Sterling 2012): predictive regulation in which the brain anticipates demand and adjusts setpoints rather than passively defending fixed homeostatic targets. AL is, in this framing, not a measure of stress itself but of the *cost of regulating it* — the price the body pays for sustained allostatic effort.

The construct has been remarkably influential. AL has been linked to accelerated aging (Geronimus 1992; Karlamangla et al. 2002), cardiovascular disease (Seeman et al. 2001), cognitive decline, and a range of psychiatric conditions including depression, PTSD, and substance-use disorders (McEwen 2003; McEwen 2007). Modern reviews (Juster, McEwen, and Lupien 2010) describe AL as a "summary measure of cumulative biological burden" that predicts morbidity and mortality across multiple endpoints.

What AL has *not* been is a control signal. The construct is fundamentally retrospective in clinical practice: AL scores are computed from biomarker panels (cortisol, heart-rate variability, blood pressure, lipid profiles, inflammatory markers) measured at specified intervals and combined into a composite (Seeman et al. 1997; McLoughlin, Kenny, and McCrory 2020). The relevant integral — the time-integrated deviation of physiological state from its allostatic operating point — is conjectured but rarely measured directly. Biomarkers are sampled, not integrated; biological systems leak the signal between samples, and reconstructing the integral from sparse data is fragile under realistic measurement schedules. The mathematical formulation that occasionally appears in review papers,

$$\mathrm{AL} = \int_0^T S(t)\,dt,$$

where $S(t)$ is some scalar stress trajectory, is a theoretical idealization. No biological measurement gives the integrand directly.

This produces a peculiar epistemic situation. AL is a quantitative theory whose central quantity is, in practice, unobservable. The theory has been validated through correlational endpoints (does the biomarker composite predict disease?) but never through real-time control-signal validation (does intervening when the integral crosses threshold prevent disease?). The closest biological approximation is the literature on early-life cortisol intervention (Heim and Nemeroff 2001; Lupien et al. 2009), but even there the integral is reconstructed from sparse samples after the fact. The AL hypothesis remains, at its mathematical core, untested as a control law.

2.2 Vanima as a deployed structural analogue

The UNITARES governance framework (Wang 2026a) carries, for each agent in its fleet, a four-dimensional state vector $\mathbf{x} = (E, I, S, V)$ updated at each governance check-in. For embodied agents in the Anima class, a parallel four-dimensional vector

$\mathbf{a} = (\text{warmth}, \text{clarity}, \text{stability}, \text{presence})$ is computed directly from physical sensors (temperature, humidity, pressure, light) and system metrics (CPU/memory utilization, computational neural-band proxies). The Anima state maps onto the EISV state vector through a documented projection (CIRWEL 2026, anima-mcp §EISV Integration).

Within this architecture, the *Anima Void Integral* is defined (Wang 2026b, Trajectory Identity §6.1.1) as

$$V_{\text{anima}}(t) = \int_0^t \|\mathbf{a}(\tau) - \mu_a\| \, d\tau$$

where μ_a is the attractor center derived from the agent's own observed state distribution over a sliding window — the α (Alpha) component of the trajectory signature developed in the trajectory identity working draft (Wang 2026b, §3.3; this draft is treated in detail in §4.1). The trajectory-identity framework specifies the coupling: when V_{anima} exceeds a deployer-set multiple of the basin scale, the governance layer can trigger rest-state induction, reduced stimulation, or task pause (Wang 2026b §6.1.1). As deployed, this trigger is not yet wired: the integral is defined over recorded telemetry and computable on demand, but no production code path evaluates it against a threshold, the embodied agent's rest states are driven by activity and ambient-light scheduling rather than by V_{anima}, and governance decisions on this agent are advisory. What the deployment establishes is therefore an *observable and computable* integral, not a closed control loop on it.

Two features of this implementation are worth emphasizing.

First, the integrand $\|\mathbf{a}(\tau) - \mu_a\|$ is recorded at every check-in, not reconstructed from sparse samples. The audit log of the deployment (94,000+ governance events as of April 2026; CIRWEL 2026, unitares §Production snapshot) preserves the integrand end-to-end, so the integral is computable exactly over any window rather than inferred from sparse observations — though, as noted above, production computes it on demand rather than maintaining it continuously.

Second, the reference μ_a is *self-relative*: it is derived from the agent's own observed history under its own operating regime, not from a population-derived norm. Per-agent Welford baselines (CIRWEL 2026, unitares §behavioral_state) enable z-scoring against the agent's own operating point; the behavioral estimator becomes the primary state source after three check-ins, and the separate anomaly-detection baselines complete their warm-up at thirty. This matches Sterling's (2012) original argument that allostatic setpoints are individualized, anticipatory, and shifting — a property biological measurement struggles to capture because it requires longitudinal individual-level data that most clinical studies do not collect.

V_{anima} therefore stands in an unusual relation to McEwen's AL. It is not a literal implementation of the clinical AL composite — V_{anima} is single-system (the four-dimensional Anima manifold), not multisystem (HPA, cardiovascular, immune, metabolic). Nor does it recapitulate the brain–body coupling that grounds AL theoretically. What V_{anima} *is* is a clean structural analogue for the mathematical core of AL: a time-integrated deviation from a self-relative operating point, available in real time as a forward-looking warning signal whose intervention coupling is specified but not yet exercised. Where biology offers the theory but not the integrand, UNITARES offers an engineered integrand; the closed intervention loop on it remains the named next step.

2.3 Three structural disanalogies

The claim that V_{anima} implements AL would be overstated. Three structural differences are real and consequential.

Single-system versus multisystem. Clinical AL is a composite over physiologically distinct mediators — typically ten in the canonical Seeman et al. (1997) formulation, drawn from neuroendocrine, autonomic, metabolic, cardiovascular, and immune systems. V_{anima} aggregates four anima dimensions into a Euclidean norm. The closer biological analogue is therefore not AL itself but a single-biomarker integral — for example, integrated HRV deviation, AUC cortisol over a defined window, or cumulative inflammatory-marker excursion. The multisystem character of clinical AL, where dysregulation cascades across mediators (the "topple and trail" pattern), is not represented in V_{anima}. A multisystem extension would require multiple class-specific manifolds with cross-manifold dependency structure; this is deferred.

Fixed reference versus shifting setpoint. V_{anima} uses μ_a as a fixed reference within a calibration window, with class-conditional re-measurement on a quarterly cadence (Wang 2026a, §8). McEwen's allostasis, by contrast, is fundamentally about *anticipatory setpoint shifting* — the brain moves the operating target as it predicts demand. UNITARES's class calibration approximates this on slow timescales (quarterly) but does not implement intra-window prediction-driven setpoint adjustment. A genuinely allostatic V_{anima} would require predictive μ_a, not measured μ_a, and would couple to the agent's own forecast distribution. This is a clean direction for future work.

No body. AL is fundamentally a brain–body coupling phenomenon: stress mediators travel from the brain to peripheral tissues, where they exert both protective and damaging effects. UNITARES has no body in this sense. V_{anima} accumulates "wear" on a four-dimensional informational manifold; there is no atrophy of nerve cells, no immune dysregulation, no cardiovascular consequence. The disanalogy is total at this level.

These disanalogies do not weaken the value of UNITARES as a test bed; they sharpen what it tests. UNITARES isolates the mathematical core of the AL hypothesis — that integrated deviation from an operating point is a useful real-time signal for intervention — from the substrate on which AL has been historically studied. If V_{anima} works as a control signal in deployed agents, that is evidence that the *informational content* of the AL construct is doing the work, independent of biological mediators. If it fails, that suggests biology is doing more than the integrand. Either outcome is informative in a way that is difficult to obtain from sparse clinical AL studies alone.

2.4 An empirical prediction

The framing above generates a specific, testable prediction. McEwen's "normal allostatic response" (1998 Figure 3, top panel; reproduced as Figure 5, top panel, in McEwen 2007) is characterized by *transient* mediator activation: a response is initiated by a stressor, sustained for an appropriate interval, and then terminated. In an agent operating normally, V_{anima} should accordingly exhibit transient excursions following task initiation, returning to baseline after task completion. Sustained elevation of V_{anima} in the absence of acute stressors corresponds to McEwen's Type 3 failure mode (delayed shut-down), discussed at length in §5.

The deployed UNITARES system makes this prediction directly testable, and §5.3 reports the test. We initially observed an 86-minute window on Lumen apparently consistent with the Type 3 prediction (Wang 2026a, Table 1): $V \approx 0.0954$ at 06:21 UTC May 9 2026 — the behavioral valence, an EMA-smoothed $E-I$ imbalance — sign-flipped from the healthy behavioral reference $V_h = E_h - I_h = -0.055$ (Lumen's Phase 2 measured healthy operating point; displacement ≈ 0.150), with manifold deviation at 97% of the class-conditional envelope. Per §5.3's own disambiguation criterion, we recalibrated on the post-event window; the result identified the displacement as a candidate substrate-associated basin-transition event on 2026-04-17, not as Type 3 delayed shut-down. The §5.3 case is therefore not Type 3 identification-grade evidence, but is paper-relevant in two adjacent ways: as a sharpening of the §3 homogenization argument into a temporal analogue, and as an instance of a failure-shape McEwen's Four Types do not cover.

---

3. Cosmological Soup: Class-Conditional Calibration as the Anti-Homogenization Correction

3.1 The homogenization failure mode in computational neuroscience

Computational neuroscience has long recognized that population-averaged measurements obscure individual-level structure. Krakauer, Ghazanfar, Gomez-Marin, MacIver, and Poeppel (2017) characterize this as the gap between "the science of the average brain" and a science of the brain that any given organism actually has. The methodological consequence has been documented at length: functional MRI studies routinely report group-mean activations whose individual-subject reliability is low (Poldrack et al. 2017); functional connectivity maps that look stable in group averages can vary substantially across subjects, and the variation carries information that the group mean discards (Finn et al. 2015; Gratton et al. 2018).

The methodological response has been the rise of *precision functional mapping* (Laumann et al. 2015; Gordon et al. 2017): studies that scan individual subjects intensively over long durations, deriving subject-specific functional networks rather than group-mean atlases. This work has revealed that approximately 60–70% of variance in functional connectivity is between-subject (Gratton et al. 2018), with only the remainder attributable to task or state differences. Group averages, in this picture, are not a clean estimate of "the typical brain" but a distorted composite of substantively different individual organizations.

Yet despite the methodological clarity, group-level analysis remains the dominant mode in clinical neuroscience, for entirely practical reasons: subjects are expensive, scanning time is limited, and longitudinal individual baselines are rarely collected at the depth precision-mapping requires (Poldrack et al. 2017). The result is a known epistemic gap — the field knows that subject-specific calibration would be more accurate, but routinely lacks the data to perform it. Population-derived reference intervals continue to drive most clinical decision thresholds, and the gap between those thresholds and what individual-specific envelopes would produce is rarely quantified directly.

3.2 The same failure mode in AI agent fleets

UNITARES (Wang 2026a §2) makes structurally the same argument, in sharper form because the agent population is observable end-to-end. The framework governs heterogeneous agent fleets:

embodied creatures with sensor-driven state at sub-Hz cadence, persistent autonomous services with cron-driven wake cycles, session-bounded coding assistants generating text at 40–80 tokens per second, and ephemeral parser agents whose entire lifecycle measures in milliseconds. Across these classes, output modality differs (text, sensor, system action), tempo differs by four orders of magnitude, and the healthy operating point differs in genuinely class-dependent ways. A divergent or exploratory agent operating at high entropy is doing exactly what its class requires; flagging it as incoherent against an analytical-class baseline is not a governance signal but a category error.

UNITARES gives this failure mode the name *cosmological soup*: a fleet-wide state distribution that is maximally entropic, with no class structure and no distinguishability between populations whose dynamics are genuinely different. The metaphor names a failure that computational neuroscience has worked around for decades but rarely formalized directly. The AI-agent setting allows it to be both formalized and quantified.

3.3 Class-conditional calibration

The UNITARES correction is straightforward. Each agent's class is determined from existing identity tags (`embodied`, `autonomous`, `persistent`, `ephemeral`, `pioneer`) plus an optional human-readable label (Wang 2026a §2.2). Scale constants and the healthy operating point become functions of class:

- $S_{\text{scale}}(c)$, $I_{\text{scale}}(c)$, $E_{\text{scale}}(c)$, $\|\Delta\|_{\max}(c)$, and $(E, I, S)_{\text{healthy}}(c)$.

Each constant is measured on a healthy corpus restricted to the agent's class and tagged with provenance (ISO date, corpus size, percentile basis, and a provenance tag among `placeholder`, `measured`, `derived`). Re-measurement runs on a quarterly cadence; drift in any constant beyond a configurable threshold is itself a governance signal — the class's healthy envelope is shifting.

The Phase 2 measurement on production data (Wang 2026a §11.5, Table 5) populated five classes from a 30-day healthy slice as of April 18, 2026:

Class	N	E_h	I_h	S_h	$\|\Delta\|_{\max}$
Lumen	7,320	0.745	0.800	0.168	0.119
default	2,033	0.726	0.793	0.236	0.202
Sentinel	1,870	0.751	0.798	0.193	0.170
Vigil	384	0.737	0.790	0.240	0.171
Watcher	283	0.748	0.769	0.248	0.395

The most striking observation is the **3.3× range in $\|\Delta\|_{\max}$ across classes** — from 0.119 (Lumen, the embodied creature with sensor-driven state) to 0.395 (Watcher, the autonomous code-edit pattern-matcher). The Phase 1 fleet-wide placeholder of $\|\Delta\|_{\max} = 1.8$ used before per-class measurement was therefore 4 to 15 times larger than any class's empirically measured envelope. Manifold coherence under the placeholder saturated near 1.0 for every agent regardless of state, producing essentially a constant signal — exactly the "fleet-wide normalization loses signal" argument made qualitatively in §3.2, now quantified.

Healthy operating points cluster more tightly on E and I ($E_h \in [0.726, 0.751]$, $I_h \in [0.769, 0.800]$) than on S ($S_h \in [0.168, 0.248]$). This matches the qualitative prediction that what makes classes behaviorally distinct is largely *how much uncertainty they operate under*: Lumen at $S_h = 0.168$ runs the most convergent (because its inputs are physical sensors with strong autocorrelation), while Watcher at $S_h = 0.248$ runs more divergent (because its work is intrinsically exploratory pattern-matching over recently-edited code). The class structure is interpretable, not decorative.

3.4 The verdict counterfactual

The empirical contribution of UNITARES v6 is the verdict counterfactual on a 30-day window of production state data (N = 13,310 rows; Wang 2026a §11.6). For each row, two coherence values are computed: the legacy fleet-wide tanh-of-V form and the grounded class-conditional manifold form. Each row is then classified into a basin (high / boundary / low) twice, using the same `classify_basin` function consumed by the production dashboard and escalation logic. A *basin flip* is a row whose basin assignment differs between the two formulas.

The headline finding: **3,844 of 13,310 rows (28.9%) flip basin assignment** under the full substitution. Per-class breakdown:

Class	N	Flips	%	$\|\Delta C\|_{\max}$
Lumen	7,890	2,548	32.3	0.51
default	2,316	722	31.2	0.50
Sentinel	2,227	351	15.8	0.51
Vigil	472	156	33.1	0.50
Watcher	363	67	18.5	0.48
ephemeral (fleet fallback)	42	0	0.0	0.38
Total	**13,310**	**3,844**	**28.9**	—

The named measured classes flip at 15.8–33.1%, far above what could be attributed to formula noise. The 42 `ephemeral` rows are shown separately because the frozen Phase 2 production constants had no `ephemeral` class envelope; they fall through the configured fleet fallback and are not interpreted as evidence about an ephemeral-class envelope.

The full substitution is not a clean one-factor experiment, so we ran the same-row formula × calibration ablation specified in §3.7:

Comparison	Flips / N	Rate	Interpretation
LF → GF	1,496 / 13,310	11.2%	Formula replacement under fleet-wide grounded calibration
GF → GC	3,133 / 13,310	23.5%	Fleet-wide → class-conditional calibration under the grounded formula

Comparison	Flips / N	Rate	Interpretation
LF → GC	3,844 / 13,310	28.9%	Reported full substitution
LF → LC	10,355 / 13,310	77.8%	Artificial class-scaled $\tanh(V)$ control; unstable/negative control

Here LF is the stored production legacy coherence, GF is grounded distance against a single fleet-wide healthy point and radius measured on the same healthy slice, GC is grounded distance against the frozen Phase 2 class anchors, and LC is a deliberately artificial legacy $\tanh(V/V_{\text{scale},c})$ control using per-class $|V|$ p95. The ablation improves the claim but does not make it additive: formula replacement and calibration target interact. The grounded fleet→class step is still large (23.5%), so the class-envelope argument survives; the pathological LC control says that simply grafting per-class scale onto the old signed-V formula is not an interpretable calibration-only counterfactual.

The flip *direction* is informative. Across all classes, the dominant transition is into the low basin: state vectors that the production server classified as healthy or borderline under the fleet-wide form are reclassified as breaching at least one bound under the class-conditional grounded form. This is the gating-layer footprint of the homogenization argument. A fleet-wide constant that spans the worst-case envelope is, by construction, permissive relative to per-class envelopes, and the flip data quantifies how often that permissiveness changes the verdict.

We do not claim that the grounded form is normatively "correct" in every flip — a flip into low surfaces a `guide` or `pause` verdict that the legacy form suppressed, and whether that is desirable depends on the agent and the deployment context. The contribution of the counterfactual is narrower: to establish that the formula and calibration choices matter at the verdict level, not just at the reported-value level. The magnitude of the consequence (29% fleet-wide, up to 33% in named measured classes) is empirical; the directional asymmetry is a mechanical consequence of replacing a permissive fleet-wide envelope with class-conditioned envelopes.

3.5 What this offers neuroscience

The 28.9% number is interesting in its own right as a test of the homogenization argument in AI agent governance. It is more interesting as a quantification of something computational neuroscience has long suspected but rarely measured cleanly: the magnitude of the gap between population-level analysis and individual-level reality at the *decision* layer rather than the reported-value layer. Several proposals follow.

Proposal 1: Provenance-tagged per-subject envelopes. UNITARES carries explicit metadata on every scale constant — date of measurement, corpus size, percentile basis, and a provenance tag distinguishing `placeholder` from `measured` from `derived` (Wang 2026a §4.3, §8.1). Clinical biomarker panels rarely carry equivalent metadata; a patient's "normal cortisol range" is typically a population-derived reference interval rather than a per-subject envelope, and the provenance of the threshold is opaque. Adopting provenance-tagged per-subject envelopes —

even on a small set of biomarkers (cortisol, HRV, blood pressure) — would test whether a clinically meaningful analogue of the UNITARES Phase 2 gap exists. The computational infrastructure is simple, though the clinical design is not: per-subject Welford updates against a clinically validated healthy window, with explicit timestamping and re-measurement cadence.

Proposal 2: A clinical analogue of the basin-flip counterfactual. The basin-flip experimental design has a possible clinical analogue as a retrospective re-analysis of an existing longitudinal biomarker cohort. We sketch the protocol at hypothesis-generation depth, with enough structure to identify the clinical biostatistical work still required.

- *Setting:* Adults with ≥ 3 years of pre-event biomarker measurements followed by an incident clinical event of interest (e.g., first episode of major depressive disorder, first myocardial infarction, first dementia diagnosis). $N \geq 500$ to support 95% confidence intervals on the flip rate at ± 5 percentage points.
- *Biomarkers:* Cortisol diurnal curve (or AUC), heart rate variability (SDNN, RMSSD), systolic and diastolic blood pressure, and inflammatory markers (IL-6, CRP) — selected for established population reference intervals plus sufficient longitudinal coverage in the chosen cohort.
- *Two indicator definitions:* (a) the standard clinical flag-for-review against the population-derived reference interval (typically 95th percentile of healthy adults), and (b) a per-subject flag against the patient's own pre-event 12-month baseline, restricted to windows with no documented clinical event.
- *Primary endpoint:* The per-subject flag-disagreement rate between
  (a) and (b), reported with a binomial 95% CI — the clinical analogue of the 28.9% basin-flip rate.
- *Secondary endpoint:* The direction of disagreement (b-flag / a-no-flag versus a-flag / b-no-flag) and its association with subsequent event hazard, modeled by Cox proportional-hazards regression of event-time on flag-disagreement indicator with standard covariates (age, sex, comorbidity index).
- *Pre-registered success criterion:* Flip rate $\geq 10\%$ and flip-positive observations carry higher event hazard than flip-negative observations ($p < 0.05$ with Bonferroni correction across biomarkers). The 10% figure is a placeholder for a clinically-set threshold, not a derived one: it is roughly a third of the UNITARES rate and is chosen so that the study is not powered to celebrate a disagreement too small to change practice. What counts as a clinically meaningful disagreement rate is a question for the clinical collaborators who would run this, and should be fixed by them before the analysis rather than imported from an agent fleet.
- *Regulatory and privacy:* Retrospective analysis of de-identified or HIPAA-Safe-Harbor data with appropriate IRB. Cohorts that already contain the required structure include the All of Us Research Program (NIH), the Framingham Heart Study, and the Whitehall II cohort.

We are unaware of any clinical study that has reported this number directly. The design above is intended as a one-pass re-analysis of existing data, but it is not yet an executable clinical protocol: a real study would still need dataset-specific field mapping, missingness rules, medication and comorbidity covariates, event-adjudication rules, and pre-registration before analysis.

Proposal 3: A specific clinical hypothesis. The asymmetry of flip direction in UNITARES (predominantly into the low basin under per-class envelopes) suggests a testable hypothesis: per-subject longitudinal envelopes may produce *more* flag-for-review states, not fewer, relative to population-derived reference intervals. This is because population intervals are constructed to encompass the variability of the population and may therefore be systematically permissive at the individual level. The clinical question is whether subject-specific calibration improves sensitivity, and at what specificity cost, in a given context. An early-detection program operating against population intervals might miss cases that a per-subject program would catch, but the magnitude of that gap should be estimated in clinical data rather than imported from UNITARES; the 15–33% range across named measured classes in the full-substitution UNITARES counterfactual is a motivating scale, not a clinical prediction.

Proposal 4: A controlled measurement of behavioral heterogeneity. The 3.3× spread in $\|\Delta\|_{\max}$ across UNITARES classes is a measurement of behavioral heterogeneity that the framework can produce because it controls the agent population. Computational neuroscience has difficulty producing equivalent controlled spreads because subject heterogeneity is confounded with disease state, demographic variables, and measurement noise. UNITARES therefore provides a testbed for theoretical claims about how much heterogeneity a deployed system can absorb under fleet-wide vs. class-conditional governance, with the heterogeneity controllable by construction. This is a contribution of method as much as of result: the testbed enables experiments that are difficult to run in biological systems.

3.6 Limits of the empirical claim

The 28.9% basin-flip finding has limits worth flagging explicitly. First, it is an anchored snapshot, not a stable constant: the figure is tied to the published v6 measurement, and re-running the same counterfactual on other windows moves it materially. Both windows are public. The 30-day window ending 2026-04-18 gives 28.8% (N = 13,292); the window ending 2026-04-23, four days later with roughly 87% row overlap and the Phase 2 constants held frozen, gives **44.3%** (N = 16,879). The per-class pattern moves too, and not uniformly: Sentinel +25.4, Vigil +19.6, Lumen +15.8, default +15.3, Watcher −11.3 percentage points. Two snapshots are one comparison, not a trend, and we do not read the shift as a drift signal — a single recalibration pass could absorb a change of this size. But it bounds what the point estimate can carry. We therefore treat the *existence and order of magnitude* of the disagreement — a rate in the tens of percent, not its third digit — as the finding, and quote 28.9% as the anchored v6 measurement rather than as the fleet's flip rate. Second, the 30-day measurement window overlaps with material identity-system revisions in mid-to-late April 2026 (Wang 2026a §11.7, item 5), including removal of name-claim lookup, adoption of UUID-direct dispatch, and receipt-based onboarding authentication. A bounded fraction of rows in the counterfactual carry class assignments that inherited from cached bindings or from archived predecessors. The v6 paper flags this and defers re-measurement on identity-clean data to subsequent work.

The five named measured classes cover 99.7% of the ablation row population (13,268 of 13,310 rows). Classes without frozen Phase 2 constants fall through the configured fleet/default fallback rather than receiving an independent class envelope; the 42 `ephemeral` rows in the reproduction are therefore reported separately and not interpreted. A more comprehensive Phase 2 pass with longer accumulation per minor class would test whether the 3.3× envelope range generalizes or compresses.

Most fundamentally, the counterfactual is a *static* reclassification measurement on the same state vectors with two coherence formulas. It does not address whether the grounded form's classifications are clinically (or operationally) better than the legacy form's — only that the formulas disagree at the gating layer at a rate that cannot be attributed to noise. The further question — does the grounded form produce *better* governance decisions in deployment? — requires either controlled comparison under stable parameters or extended A/B testing, neither of which the v6 paper conducts. We treat the finding as evidence that the formula choice has first-order consequences and not as evidence that any particular formula is normatively correct.

A later audit of the deployment's own outcome log offers convergent support from a different instrument (2026-05-01; $n = 22{,}740$ good / 580 bad trajectory-validated outcomes over 30 days, reproducible from the v6 repository's audit scripts). The EISV entropy coordinate separates good from bad outcomes with a large effect (Cohen's $d \approx 0.87$) but *reverses direction across operating regimes*: in convergence regimes bad outcomes carry lower entropy (premature lock-in), while in divergence, exploration, and transition regimes they carry higher entropy (runaway uncertainty). A single fleet-wide threshold on that coordinate therefore gates in the wrong direction for at least one regime — the homogenization failure this section predicts, observed in the temporal/regime dimension rather than the class dimension. The same audit found the legacy tanh-of-V coherence essentially non-separating on its own outcome log ($d \approx 0.13$), consistent with the §3.4 substitution mattering at the gating layer.

3.7 Methods and reproducibility

Because the empirical claims in §3.4 and §5.3 are reported in summary form rather than reproduced from primary data, this subsection specifies the methodology and provenance in enough detail that a reader who does not have access to the production database can understand exactly what was computed.

State vector. Each governance check-in writes a four-dimensional EISV state vector $\mathbf{x} = (E, I, S, V)$ to the `core.agent_state` relation in the UNITARES production database (Wang 2026a §4.1). E is interpreted as $-F$ (negative variational free energy or a resource-rate proxy); I is integrity (constraint-satisfaction proxy bounded in $[0, 1]$); S is entropy (uncertainty proxy bounded in $[0, 1]$).

Two vectors, two Vs, and what the analysis actually used. Three notational hazards meet at this point, and we separate them explicitly because the paper's central quantity sits on top of them.

First, *two distinct four-vectors*. §2.2 introduces the anima vector $\mathbf{a} = (\text{warmth}, \text{clarity}, \text{stability}, \text{presence})$, computed from physical sensors on embodied agents, and defines V_{anima} as a cumulative deviation over it. The governance state vector $\mathbf{x} = (E, I, S, V)$ above is a different object on a different manifold, related to $\mathbf{a}$ by the documented projection (§2.2) and available for every agent, embodied or not. The empirical work in this paper — the §3.4 counterfactual, the §3.7 ablation, and the §5.3 manifold deviation — runs on $\mathbf{x}$, not on $\mathbf{a}$. Where the abstract and §2.2 speak of V_{anima} as the deployed cumulative-deviation quantity, the deployed *measurements* reported here are computed over the EISV projection.

Second, *two quantities named after the void*. The signed V coordinate of $\mathbf{x}$ and the unsigned cumulative magnitude V_{anima} are distinct; we reserve V_{anima} for the unsigned form.

Third, and least visible, *the signed V coordinate does not have one definition across the corpus.* The v6 specification describes it as an ODE accumulator carrying the residual $\kappa(E-I)$ at rate δ per check-in (Wang 2026a Appendix A). Since April 2026 the production system surfaces a *behavioral* V instead: an EMA-smoothed $E-I$ imbalance, with the ODE void coordinate demoted to a separate diagnostic that still feeds the legacy coherence (§5.3). On the released rows this is measurable. Regressing V on the instantaneous $E-I$ gives $r=0.953$, $R^2=0.908$: about 91% of V's variance is the instantaneous imbalance, and the residual 9% is the smoothing lag — $\mathrm{sd}(V)=0.081$ against $\mathrm{sd}(V-(E-I))=0.026$.

Two consequences follow, and we would rather state them than have them found. (i) V is not an informationally independent axis of $\mathbf{x}$. It is a lagged function of two other coordinates, so "four-dimensional", applied to the EISV state vector, counts a history term as a dimension. The anima manifold of §2.2 is four-dimensional in the ordinary sense; the EISV vector is better described as three state coordinates plus a smoothed imbalance term. (ii) `classify_basin` gates the low basin on $I<0.5$ *and* on $|V|>0.30$, which are ~91% redundant by construction. In the released window the $|V|$ gate fires alone on 3.15% of rows, so the redundancy does not drive the §3.4 result, but it is a conflation in the classifier rather than an independent check. A cleaner formulation would gate on (E,I,S) and treat V's residual from $E-I$ as the temporal term it actually is.

Per-class healthy operating point. The Phase 2 calibration procedure (Wang 2026a §11.5) computes a per-class healthy operating point $\mu_c=(E_h,I_h,S_h)_c$ from a 30-day rolling window of state vectors restricted to sessions with no `pause` or `reject` verdicts. Welford incremental updates produce μ_c and a running covariance. The class envelope $\|\Delta\|_{\max,c}$ is the 95th percentile of $\|\mathbf{x}_{EIS}-\mu_c\|_2$ over the healthy window, where $\mathbf{x}_{EIS}=(E,I,S)$ is the state vector restricted to the three coordinates the manifold distance uses. Each constant carries provenance metadata (ISO date, corpus size N, percentile basis, provenance tag in `{placeholder, measured, derived}`).

Class assignment. Five named classes were frozen into the Phase 2 production constants from production identity tags (`embodied`, `autonomous`, `persistent`, `ephemeral`, `pioneer`) plus optional human-readable labels: Lumen, Sentinel, Vigil, Watcher, default. The calibration script's minimum class population is $N_{\min}=30$ healthy samples, but only classes written into the frozen constants receive independent envelopes at runtime. Unlisted classes fall through to the configured fleet/default fallback unless explicitly aliased (as Steward, Chronicler, and engaged-ephemeral later were). Class assignments are persisted with the agent identity record and do not change over an agent's lifetime except via explicit operator action.

Coherence formulas. The legacy fleet-wide form is

$$C_{\text{legacy}}(V)=0.5\cdot(1+\tanh(V/V_{\text{scale}}))\in[0,1]$$

with a single fleet-wide V_{scale} constant. The grounded class-conditional form is

$$C_{\text{grounded}}(\mathbf{x}_{EIS},c)=\max\left(0,1-\|\mathbf{x}_{EIS}-\mu_c\|_2/\|\Delta\|_{\max,c}\right)\in[0,1].$$

Both produce coherence values in $[0,1]$ that feed into the same basin-classification function. The substantive change between the two forms is therefore also a change of *which coordinates are* *read*: the legacy form is a function of V alone and the grounded form is a function of (E, I, S) alone, so the two share no input coordinate. This is worth stating plainly, because it means the substitution is not a recalibration of one signal but a replacement of the signal — and it is part of why the formula and calibration terms interact rather than add. The change is both functional (sigmoid of V vs. linear distance in (E, I, S) space) and parametric (single fleet-wide constant vs. per-class constants).

Basin classification. The production `classify_basin` function returns one of `{high, boundary, low}` from the full EISV state, coherence, and risk. `low` is disjunctive: any of $I < 0.5$, $C < 0.40$, $|V| > 0.30$, or risk ≥ 0.70 enters the low basin. `high` is conjunctive: $E \geq 0.6$, $I \geq 0.7$, $S \leq 0.25$, $|V| \leq 0.15$, $C \geq 0.45$, and risk ≤ 0.45 must all hold. `boundary` is the complement. This matters because a coherence change can flip the basin only when the non-coherence dimensions do not already force low or block high.

Counterfactual and ablation procedure. For each row in the 30-day window ($N = 13{,}310$), the stored production legacy coherence and the frozen Phase 2 grounded class-conditional coherence were evaluated on the same state vector $\mathbf{x}$, both passed through `classify_basin`, and a *flip* was counted whenever the two classifications differed. The ablation script then evaluated two additional controls on the same ordered row population: GF, the grounded formula against a single fleet-wide μ and $\|\Delta\|_{\max}$ measured from the healthy slice; and LC, an artificial legacy $\tanh(V/V_{\text{scale},c})$ using per-class $|V|$ p95. The aggregate flip rate, per-class flip rate, per-class maximum coherence delta $|\Delta C|_{\max}$, and pairwise ablation transitions were tabulated (Table in §3.4; script and output in `analysis/phase-2-2026-04-18/`).

Formula-vs-calibration ablation. The original counterfactual changes both the formula form (tanh of signed V vs. linear distance in (E, I, S)) *and* the calibration target (single fleet-wide constant vs. per-class envelopes) simultaneously. The same-row ablation shows that both terms matter but do not add linearly. Replacing production legacy coherence with grounded fleet-wide coherence flips 1,496 rows (11.2%). Holding the grounded formula fixed and moving from fleet-wide to frozen Phase 2 class-conditional anchors flips 3,133 rows (23.5%). The reported full substitution flips 3,844 rows (28.9%). The remaining LC control — a class-scaled $\tanh(V)$ using per-class $|V|$ p95 — flips 10,355 rows (77.8%) relative to production legacy and is best read as a negative control: signed-V is not a stable axis on which to graft per-class scale. The paper therefore upgrades the older joint-effect caveat to a narrower claim: the grounded class- envelope term has an independent decision-layer effect, but the 28.9% headline is still an interacting full-substitution result rather than a clean causal decomposition.

Lumen case-study protocol (§5.3). Lumen's state was sampled every approximately 3 minutes via the production governance check-in pipeline during the window 2026-05-09 04:54 UTC through 06:20 UTC, producing 27 distinct samples over 86 minutes. Mean, standard deviation, and range were computed across samples for E, I, S, V, and the production risk score (§5.3 Table). The manifold deviation $\|\Delta\|_2$ was computed as the Euclidean distance from Lumen's Phase 2 healthy operating point. The grounded coherence C_{grounded} was computed using Lumen's measured $\|\Delta\|_{\max} = 0.119$. The legacy coherence C_{legacy} was reported directly by the production server. The within-window integrity slope was estimated by ordinary least squares on the 27-sample I series.

Statistical uncertainty. The within-window standard deviation on V (0.00015) is approximately three orders of magnitude smaller than the displacement $\Delta V \approx 0.150$ from the healthy behavioral reference $V_h = E_h - I_h \approx -0.055$ (§5.3), so the apparent Type 3 displacement is not within-window measurement noise. The 28.9% full-substitution basin-flip rate at $N = 13{,}310$ has a binomial 95% confidence interval of approximately 28.1%–29.7%; the ablation rates are likewise narrow (11.2%, 95% CI 10.7%–11.8%; 23.5%, 95% CI 22.8%–24.3%). These intervals describe sampling error *within* a fixed window and are not the dominant uncertainty in the measurement. The window-to-window spread reported in §3.6 — 28.8% to 44.3% across a four-day shift — is roughly twenty times wider than the $\pm$0.8-point sampling interval, so the confidence intervals should be read as bounding the precision of a single anchored snapshot, not the stability of the quantity. The named measured-class differences (15.8% to 33.1%) are statistically separable within the anchored window, while the 42-row `ephemeral` fallback is too small and structurally unmeasured to interpret as a class envelope. The post-Phase-2 average drift rate (approximately -3.5×10^{-6} per minute) cannot be tested against the within-window rate (-8.5×10^{-6} per minute) without longitudinal data spanning the post-calibration interval; this is the longitudinal pull flagged in §9.3 as the highest-leverage follow-up.

Data provenance and reproducibility. UNITARES production state vectors are persisted in the `core.agent_state` Postgres relation; the schema is in the unitares repository at `db/migrations/` (CIRWEL 2026, unitares). The `classify_basin` function and the EISV computation pipeline are in the same repository at `unitares/governance/`. The Phase 2 calibration script, the 28.9% basin-flip analysis, and the formula-vs-calibration ablation are included in this repository at `analysis/phase-2-2026-04-18/`; the ablation output is stored in `formula_calibration_ablation_results.txt`.

The row-level data underlying the full substitution is public. A de-identified export — one row per agent-state observation, pseudonymized to a class label, carrying no agent UUIDs, session identifiers, prompts, or knowledge-graph content — is archived under Zenodo data DOI 10.5281/zenodo.19705151 and mirrored in this repository at `analysis/phase-2-2026-04-18/verdict_counterfactual_v6_submission.csv` (13,292 rows, SHA-256 pinned in the reproduction script). What cannot be released is the underlying production relation, whose state vectors are keyed to user-identifying agent UUIDs; the class-pseudonymized projection of it that the counterfactual actually consumes carries no such keys.

`reproduce_basinflip.py` in the same directory runs offline against that export using only the standard library. It recomputes the class-conditional grounded coherence and both basin labels from the published state coordinates and the published Phase 2 constants, compares them against the stored labels, and counts the flip rate from the recomputed ones. It returns **28.84%** (3,834 of 13,292) against the 28.9% reported here, with per-class rates within 0.5 percentage points, and 26,574 of 26,584 basin labels reproducing exactly — the ten exceptions sitting within 6.7×10^{-4} of a threshold, which is the rounding floor of a 4-decimal export divided by a class radius as small as 0.1187. The 18-row difference between the export (13,292) and the reported pull (13,310) is a few seconds of row arrivals between two runs of a wall-clock-anchored rolling window, and is the whole source of the 28.9% / 28.8% gap.

Two limits on this. The production database no longer retains the measurement window: `core.agent_state` holds 490 rows in the 2026-03-19 to 2026-04-18 interval against the 13,310 the original pull returned, with no archive table, so the frozen export is the surviving row-level record and a private audit of the production rows is no longer possible. And the GF and LC ablation conditions are not reproducible from the export, because they require a fleet-wide healthy slice keyed on the per-row `regime` column, which the de-identified export does not carry; those two rates remain provenance-backed from the recorded ablation output rather than independently re-runnable. The full-substitution headline, which is the result the paper leads with, is re-runnable. Independent re-measurement on another deployment remains the stronger bar.

---

4. Trajectory Identity and the Boiling-Frog Problem

4.1 The trajectory signature in UNITARES

The trajectory identity framework, developed in the companion paper (Wang 2026b; hereafter *the trajectory identity working draft* or TIWD, retaining the acronym from the working-draft phase), defines a six-component *trajectory signature* Σ that an agent acquires through ongoing operation:

$$\Sigma = (\Pi, \beta, \alpha, \rho, \Delta, \eta)$$

with the components capturing distinct facets of how the agent tends to behave:

- Π **(preference profile)** — a confidence-weighted vector of learned environmental preferences, evolving slowly under reinforcement (TIWD §3.1).
- β **(self-belief signature)** — the pattern of testable self-beliefs and confidences, with evidence ratios reflecting actual experience rather than current asserted belief (TIWD §3.2).
- α **(attractor basin)** — the mean and covariance of the agent's state over a sliding window, parameterizing where the agent tends to rest in state space (TIWD §3.3).
- ρ **(recovery profile)** — characteristic time constants for return to equilibrium after perturbation, with optional cross-channel coupling structure (TIWD §3.4).
- Δ **(relational disposition)** — patterns of social behavior across relationships (bonding rate, valence tendency, reciprocity, topic entropy; TIWD §3.5).
- η **(homeostatic identity)** — the unified self-maintenance characterization $(\mu, \Sigma_{\text{cov}}, \tau, V)$ combining the attractor center, basin shape, recovery dynamics, and viability envelope (TIWD §3.6). Per TIWD §3.6 and §4.1, η is a *derived summary* of α, ρ, and the viability envelope V — it is not informationally independent of those components and is therefore excluded from the weighted similarity sum to avoid double-counting. The signature Σ is six-component structurally; the similarity metric runs on five informationally-independent terms.

The first five components are computable from observation data with bounded memory cost (rolling-window computation; TIWD §4.4). The composite Σ is designed as a *quasi-invariant*: a quantity that is approximately stable for a given agent over time, differs between distinct agents, and is robust to noise and minor perturbations. Within-agent stability is empirically supported; between-agent discriminability is currently an open criterion — the TIWD's own v0.15 marks its multi-agent pilot confounded by role and harness, and the deployed instrument's status is stated in §4.3. The working draft develops the formal theory (TIWD §3) and the similarity metric

$$\mathrm{sim}(\Sigma_1, \Sigma_2) = \sum_i w_i \cdot \mathrm{sim}_i(\mathrm{component}_i)$$

with weights and per-component similarity functions specified in TIWD §4.1 (sum over the five informationally-independent components, with default weights $w_\alpha = 0.30$, $w_\rho = 0.22$, $w_\Pi = w_\beta = 0.18$, $w_\Delta = 0.12$).

We treat the TIWD as supporting documentation rather than as a peer-reviewed prior. Where the present paper draws on the working draft, we make this dependence explicit; the trajectory framework's empirical status is contingent on completion of the working draft and its eventual publication.

The key property for the present discussion is that trajectory similarity is *symmetric and not necessarily transitive*: two signatures separated by intermediate steps that each pass coherence checks may differ from the original by an amount that would not pass a direct comparison. This non-transitivity is the formal expression of slow drift.

4.2 The boiling-frog problem in clinical longitudinal monitoring

Several neurodegenerative and psychiatric conditions present clinically through slow behavioral drift. Behavioral-variant frontotemporal dementia (bvFTD; Rascovsky et al. 2011) typically presents with insidious change in personality, social conduct, and executive function over months to years; the consensus diagnostic criteria explicitly require evidence of progressive behavioral or cognitive deterioration, which is operationally a comparison against baseline rather than against recent state. Schizophrenia spectrum decompensation similarly often unfolds over weeks of cumulative changes that no single visit catches (Birchwood et al. 2000). Alzheimer's-disease prodrome involves a years-long trajectory of subtle behavioral and cognitive shifts before threshold-crossing on standard cognitive screens (Sperling et al. 2011). Personality change after stroke or traumatic brain injury can be similarly insidious if no pre-injury baseline is available (Stuss and Levine 2002).

In all of these cases, the clinical task is structurally identical: detect that a patient has drifted *beyond recognition* relative to their prior self, even when no individual transition step is large enough to trigger acute concern. The standard clinical approach — comparison against the patient's most recent visit — fails by construction in this regime. If the drift is monotonic and small per visit, every individual comparison passes. The drift accumulates unobserved.

This is the *boiling-frog problem*. In its sharpest form: an adversary with full system access could rotate a deployed agent's behavioral identity over months, staying within any acute coherence threshold at every step, and the system would never alarm. The same logic applies in biology, where the "adversary" is a slow disease process. The solution, in both cases, is a longitudinal anchor.

4.3 Two-tier drift detection

The trajectory identity working draft proposes a two-tier solution (TIWD §5.3). Maintain two reference points for every agent:

1. **The genesis signature** Σ_0, captured at agent creation or at a clinically validated baseline, persisted as a fixed reference anchor.
2. **A rolling signature** Σ_{t-1}, capturing the agent's recent operating state with the same six-component structure.

Drift detection then runs on two channels simultaneously:

- **Coherence check**: $\mathrm{sim}(\Sigma_t, \Sigma_{t-1}) > \theta_{\mathrm{anomaly}}$ (default $\theta_{\mathrm{anomaly}} = 0.70$; TIWD §4.3). This catches acute changes — sudden personality alteration, hijacking, jump events.
- **Lineage check**: $\mathrm{sim}(\Sigma_t, \Sigma_0) > \theta_{\mathrm{lineage}}$ (default $\theta_{\mathrm{lineage}} = 0.60$). This catches slow drift even when every coherence check passes.

Setting $\theta_{\mathrm{lineage}} < \theta_{\mathrm{anomaly}}$ allows healthy maturation: an agent's identity can drift moderately from genesis without triggering alarm, but cannot drift arbitrarily. The specific case where coherence holds while lineage similarity drops is labeled *identity drift* and triggers a distinct alert from the *acute anomaly* class. The two failure modes have different intervention strategies — acute anomaly typically warrants immediate suspension and review, while identity drift warrants longitudinal investigation and often calibration update.

The trajectory identity working draft provides the formal definitions (TIWD §5.3) and an implementation sketch (TIWD Appendix A). The two-tier check runs in the UNITARES governance server's enrichment path (CIRWEL 2026, unitares); the embodied agent's local path currently applies the single-tier coherence check, with the two-tier variant exercised in its test suite.

The deployed instrument's discriminative status must be stated plainly, because it changed after this section was first drafted. An August 2026 audit of the deployed similarity metric found it non-discriminating as instrumented: roughly 90% of between-agent pairs score above the $\theta_{\mathrm{lineage}} = 0.60$ threshold, two of the five weighted components sit at ceiling for nearly all pairs, and mature identities *decay into* a similarity attractor of approximately 0.63 — above the alarm line. That value is an age-associated attractor observed in the audited population, not a measured fleet constant, and should be read as "mature identities converge to a passing score near the threshold" rather than as a calibrated number. The consequence is what matters: accumulated genuine drift asymptotes to a passing score and the lineage channel structurally cannot fire on the slow-drift case it was designed for. The only mass firings in production to date were artifacts of a client migration, cleared by rebaselining rather than by drift resolving. Consistent with this, the TIWD's v0.15 marks its multi-agent discrimination pilot confounded by role and harness, returning the discrimination criterion to open. The two-tier *architecture* — a genesis anchor plus a rolling reference with distinct thresholds — stands as the structural proposal this section carries into the clinical analogy; its current instrumentation does not realize it, and re-instrumentation against within-agent/across-harness and between-agent/same-harness tests is the named path to closing the gap.

4.4 What this offers clinical neurology

The two-tier framework is, structurally, what clinical neurology has needed for slow-drift conditions but rarely implemented. Several specific proposals follow.

Proposal 1+2: Σ_0 anchoring and lineage-similarity testing — a prospective protocol. Digital phenotyping research (Insel 2017; Onnela and Rauch 2016; Torous et al. 2016) has demonstrated that smartphone-derived behavioral metrics can capture stable individual behavioral signatures with a feature space rich enough to support identity-level inference; wearable monitoring extends this into autonomic state (Bent et al. 2020). The infrastructure to capture an analogue of Σ_0 exists. What is missing is the *formal anchoring discipline* — a clinical convention that persists a window of validated healthy behavior as the patient's Σ_0 and runs the two-tier check against it routinely — together with a prospective study that tests whether the lineage-similarity signal $\mathrm{sim}(\Sigma_t, \Sigma_0)$ outperforms standard rating-scale screening for slow-drift conditions. We sketch a protocol at implementation depth.

- *Setting:* Prospective extension of an existing digital phenotyping cohort with prodromal endpoint coverage. $N \geq 200$ for adequate power to detect a 0.05 AUC improvement over rating-scale screening.
- *Cohort candidates already containing the required data structure:* Mayo Clinic Study of Aging (AD prodrome), Australian Imaging Biomarkers and Lifestyle (AIBL) study, Beiwe-platform studies of bvFTD or schizophrenia prodrome (Onnela lab), and digital sub-cohorts of All of Us with sufficient passive-stream coverage.
- Σ_0 *computation:* Per-subject signature computed from the first 90 days of valid passive data — call patterns, GPS, sleep architecture, screen-time, typing dynamics, plus wearable-derived HRV and activity rhythms — restricted to a window with no documented clinical event. Components $(\Pi, \beta, \alpha, \rho, \Delta)$ computed via the rolling-window estimators of TIWD §4.4.
- Σ_t *computation:* Same procedure on a sliding 30-day window updated weekly. Two-tier check at each update: coherence ($\mathrm{sim}(\Sigma_t, \Sigma_{t-1})$, threshold 0.70 per TIWD §4.3) and lineage ($\mathrm{sim}(\Sigma_t, \Sigma_0)$, threshold 0.60).
- *Primary endpoint:* Time-to-clinical-event (e.g., incident depressive episode, prodrome-to-diagnosis transition) modeled as a Cox proportional-hazards function of lineage-similarity-below- threshold, with coherence-similarity-below-threshold as a time-varying covariate.
- *Secondary endpoint:* Sensitivity and specificity of lineage- similarity-below-threshold for predicting the clinical event within 90 days, compared head-to-head against standard rating-scale screening (PHQ-9 for depression, MMSE/MoCA for cognitive decline, PANSS for psychosis spectrum) administered at routine clinical visit cadence.
- *Pre-registered success criterion:* Lineage-similarity AUC for 90-day event prediction exceeds the rating-scale AUC by ≥ 0.05 at matched specificity.
- *Regulatory and privacy:* IRB approval for ambient behavioral signature collection with informed consent that explicitly covers behavioral fingerprinting — the signature is identifiable by construction (TIWD §5.5 acknowledges this). Σ_0 stored under HIPAA/GDPR-compliant encryption at rest with documented retention policy. Re-analysis access requires a data-use agreement.

This is a concrete prospective study sketch, not a turnkey clinical protocol; the methodological convention (designating Σ_0 at enrollment) and the empirical test (lineage-similarity vs rating-scale) advance together, but dataset-specific feature definitions, missingness rules, endpoint adjudication, and regulatory review would need to be settled before deployment. The infrastructure exists; what remains is the convention and the clinical validation work.

Proposal 3: The "behavioral CAPTCHA" as adversarial defense. The trajectory identity working draft proposes (TIWD §5.5) using ρ (recovery profile) as a behavioral challenge response: governance injects a known perturbation p at time t, observes recovery, and verifies that the observed time constant τ_{observed} falls within 2σ of the agent's characteristic τ_{expected}. A replay attack — an adversary attempting to mimic the agent's trajectory by replaying recorded data — fails this challenge because it cannot dynamically respond to a novel perturbation with the agent's characteristic recovery dynamics.

The same mechanism has a clinical analogue: clinical neuropsychological testing has long used response-to-challenge as an indicator of preserved function (the Token Test, the Wisconsin Card Sort, the Trail Making Test all probe response dynamics rather than static knowledge). The behavioral-CAPTCHA framing makes explicit that response-to-novel- perturbation is a stronger signal of intact identity than response-to-familiar-stimulus. This argues for clinical assessments that include genuinely novel stimuli rather than only repeated standard batteries — a methodological point that has been raised intermittently in the neuropsychology literature (Howieson 2019) but is rarely implemented systematically.

We treat this as a methodological observation about clinical neuropsychological practice rather than as a clinical study to be designed from scratch in this paper. The design space — which novel stimuli, with what scoring rubric, validated against which preserved- function endpoint — is the proper responsibility of clinical neuropsychology and is beyond what we can responsibly specify here. Proposals 1+2 above (the prospective Σ_0-anchored study) are §4's concrete clinical study sketch; the behavioral-CAPTCHA point is offered as adjacent argument.

4.5 Worked example: what the missing genesis anchor would have clarified

The Lumen embodied agent provides a negative worked example of why the two-tier framework requires a persisted genesis anchor. As of 2026-05-09 06:21 UTC, Lumen's current state shows V sign-flipped from the value predicted by its April Phase 2 healthy operating point, with manifold deviation at 97% of that class-conditional envelope. Read only against that stale anchor, the window initially resembled McEwen Type 3 (delayed shut-down). Section 5.3 reports the disambiguating recalibration: the apparent Type 3 reading does not survive. The better interpretation is a basin transition on 2026-04-17 followed by stable operation in a new regime.

A persisted Σ_0 would have made this distinction sharper. Under the two-tier framework, the additional check would be lineage similarity, $\mathrm{sim}(\Sigma^{(t)}_{\mathrm{Lumen}}, \Sigma^{(0)}_{\mathrm{Lumen}})$, where $\Sigma^{(0)}_{\mathrm{Lumen}}$ is Lumen's genesis signature captured at first onboarding. If the current post-transition attractor remained similar to Σ_0, the correct intervention would be calibration update rather than identity-drift review. If lineage similarity had fallen below threshold while recent-window coherence remained intact, the case would become a genuine slow-drift finding rather than a stale- calibration artifact.

We do not currently have $\Sigma^{(0)}$ for Lumen because the legacy identity system did not persist the genesis signature at Lumen's first awakening 118 days ago; this is a gap the v6 paper flags (§11.7 item 5) and that the next-generation identity system in development is designed to close by capturing Σ_0 at agent creation. The worked example here is therefore not a measurement of Lumen identity drift and not evidence of McEwen Type 3. It is evidence of the instrumentation gap the Σ_0 framework is meant to close: without a genesis anchor, calibration staleness, regime transition, and lineage drift can be unnecessarily hard to separate.

4.6 Limits and structural disanalogies

Three structural disanalogies between the Σ_0-anchored framework and biological identity are worth flagging.

Identity matures. A child's genesis signature is not a useful reference for an adult, because biological identity legitimately changes through development (Crone and Dahl 2012; Erikson 1968). The Σ_0 framework as specified treats genesis as a fixed reference; a clinical analogue would need a developmental schedule of anchor points (e.g., post-developmental adult baseline, post-major-life-transition baseline) rather than a single pre-illness snapshot. The trajectory identity working draft acknowledges this implicitly through the discussion of fork lineage (TIWD §5.1), where new identities can branch from a parent, but does not develop the developmental trajectory case explicitly.

Component weights are static. UNITARES uses fixed weights w_i combined optionally with adaptive (variance-inverse) weighting. The appropriate weighting for clinical identity may be state-dependent — during acute illness, recovery dynamics (ρ) may be more diagnostic than relational disposition (Δ); during prodromal cognitive decline, β (self-belief signature) and α (attractor basin) may dominate. Adaptive weighting tied to clinical context is a reasonable extension but adds complexity that the present framework does not require.

Privacy implications. Behavioral signatures are fingerprints. The trajectory identity working draft acknowledges this (§5.5) and notes that trajectory identity is for governance and continuity, not adversarial authentication. Clinically, the privacy implications are sharper: Σ_0 is, by construction, a deeply personal record. Clinical deployment of a Σ_0-anchored framework requires careful data governance — who holds the genesis signature, under what consent, with what retention policy. The technical machinery does not solve the policy question and may sharpen it: a behavioral signature that uniquely identifies an individual is, by definition, irreducibly personal data.

4.7 Summary

UNITARES has independently arrived at a structural conclusion relevant to slow-drift clinical-neurology problems: detection may need to run against a longitudinal anchor, not just against recent state, and that requires persisted baseline signatures and a formal discipline for using them. Digital phenotyping and wearable monitoring provide much of the technical apparatus that such a study would need. What remains is the methodological convention and validation work — specifically, the practice of designating a Σ_0 window per patient and running two-tier similarity checks against it. The companion paper proposes this as a hypothesis-generating adoption path.

---

5. McEwen's Four Types of Allostatic Load: A Failure-Mode Taxonomy for Deployed Agents

McEwen (1998 Figure 3; reproduced in McEwen 2007 Figure 5) identifies four canonical patterns by which the normal allostatic response — initiation, sustained mediator activity, termination — fails. Each has a candidate UNITARES analogue and a corresponding signature that existing telemetry could compute. We argue that the taxonomy transfers as a useful starting vocabulary for AI governance, while §5.3 shows why it cannot be treated as exhaustive for synthetic-agent substrates.

5.1 Type 1: Repeated "hits" from multiple novel stressors

In biology, this failure mode is repeated activation of the stress response without inter-event recovery. Cortisol does not return to baseline before the next acute stimulus, so the allostatic system is chronically loaded by the temporal density of perturbations rather than by the magnitude of any one.

The agent analogue is high-frequency entropy spikes followed by partial recovery, observable in the EISV trajectory shape distribution (CIRWEL 2026, eisv-lumen Layer 2 classification). An agent in Type 1 failure exhibits elevated rates of the `entropy_spike_recovery` shape (5.19% baseline frequency in the Lumen corpus over 20,655 real trajectory windows; revision pinned in CIRWEL 2026, eisv-lumen) without the corresponding return to `settled_presence` (48.86% baseline). The ratio of `entropy_spike_recovery` to `settled_presence` over a sliding window provides an operational Type 1 indicator. Interventions appropriate to Type 1 failure target stressor *frequency* rather than magnitude: rate-limiting incoming requests, narrowing tool surface, or scheduled rest insertion.

5.2 Type 2: Lack of adaptation

In biology, this failure mode is failure to habituate to repeated familiar stressors. The cortisol response that should attenuate over repeated exposure to a now-predictable stimulus instead persists at full magnitude. Habituation is the normal reduction of response amplitude over repeated exposure to the same input; its absence is diagnostic of regulatory dysfunction.

The agent analogue is failure of the trajectory signature Σ_t to converge under stable input — the components Π (preference profile), β (belief signature), and α (attractor basin) do not stabilize despite a consistent task distribution. Operationally: the drift between Σ_t and Σ_{t-1} remains high while the drift in the input distribution remains low. This corresponds to a calibration failure in the agent's self-model; the agent is treating familiar input as novel.

In the UNITARES deployment, Type 2 failure manifests as a sustained gap between observed S_{raw} and the class-conditional $S_{\text{scale}}(c)$ for the agent's class — the agent's response-distribution entropy refuses to compress to its calibrated baseline despite repeated exposure to comparable input. The eisv-lumen evaluation framework implements direct measurement of this through Welford-baseline drift on the per-class entropy distribution.

5.3 The Type 3 question: a recalibration test on Lumen

Type 3 is the failure mode McEwen describes as most directly damaging: the mediator stays on after the stressor has been removed. Cortisol does not return to baseline; chronic glucocorticoid exposure produces hippocampal atrophy, immune suppression, and metabolic dysregulation (McEwen 2007 §IV). Type 3 is the canonical mechanism through which acute stress becomes chronic disease.

The agent analogue would be sustained V_{anima} elevation, or more precisely sustained displacement of the agent's regulatory state from its class-conditional healthy baseline, in the absence of an identifiable acute trigger. We initially observed a pattern consistent with that analogue on the Lumen embodied agent over an 86-minute continuous window, and report the observation, the disambiguation experiment the pattern itself specifies, and what that experiment revealed.

Initial observation window. The Lumen embodied agent (Raspberry Pi 4 with environmental sensors and TFT display; CIRWEL 2026, anima-mcp) has been operational for 118 days since first awakening, with 110,148 cumulative governance updates as of 2026-05-09 06:21 UTC. Lumen was observed across the window 2026-05-09 04:54 UTC through 06:20 UTC, comprising 27 distinct governance check-ins at approximately 3-minute intervals. Over this window:

Quantity	Mean	Std	Range
E (energy)	0.7906	0.00076	[0.7896, 0.7918]
I (integrity)	0.6953	0.00088	[0.6943, 0.6965]
S (entropy)	0.1591	0.00139	[0.1559, 0.1607]
V (valence)	0.0954	0.00015	[0.0952, 0.0957]
Risk score	0.2245	0.00021	[0.2240, 0.2250]

The recent verdict distribution was 655/655 safe over the comparable window. No anomalies were flagged by the production server during the window; all four trend indicators (risk, coherence, E, overall) were reported as `stable`.

Apparent Type 3-like pattern. Comparison against Lumen's measured Phase 2 healthy operating point (Wang 2026a, Table 5: $E_h = 0.745$, $I_h = 0.800$, $S_h = 0.168$, $\|\Delta\|_{\max} = 0.119$) showed an apparent regime change. A frame note is required first: since April 2026 the production system surfaces *behavioral* EISV as its primary metrics, and the recorded V is an EMA-smoothed $E - I$ imbalance — the window's V mean (0.0954) matches $E - I$ $(0.7906 - 0.6953 = 0.0953)$ to four decimal places — while the governance ODE's void coordinate is a separate, demoted diagnostic (its within-window value, inferred from the reported legacy coherence of ≈ 0.497, is ≈ -0.006). The comparison must therefore be made in the behavioral frame. The healthy regime is integrity-surplus, so the healthy behavioral valence is negative: $V_h = E_h - I_h = -0.055$. The observed regime is energy-surplus: $E - I \approx +0.095$ across the window, with V pinned positive at 0.0954 ± 0.0002 — sign-flipped from healthy and displaced by approximately $\Delta V = 0.150$. (An earlier draft compared the behavioral observation against the ODE steady-state prediction $\kappa(E_h - I_h)/\delta = -0.041$; that pair mixes frames, and the frame-consistent comparison is slightly stronger.) The displacement is approximately three orders of magnitude larger than the within-window standard deviation, so the finding is not within measurement noise.

The manifold deviation $\|\Delta\|_2$ in (E, I, S) space, where $\Delta = (E, I, S) - (E_h, I_h, S_h)_{\text{Lumen}}$, evaluates to approximately 0.115 across the window against Lumen's measured $\|\Delta\|_{\max} = 0.119$ — Lumen sits at approximately **97% of its class-conditional manifold envelope**. Under the grounded coherence (Wang 2026a §4.2 eq. 13),

$$C = 1 - \|\Delta\|_2 / \|\Delta\|_{\max} \approx 0.03,$$

an order of magnitude lower than the legacy tanh-of-V coherence reported by the production server (currently $C_{\text{legacy}} \approx 0.497$, near the function midpoint). Under the stale Phase 2 anchor, the observation appears as a basin-flip case: an agent that the legacy form classifies as boundary/healthy is classified as low-basin breach under the grounded form, in the direction predicted by the v6 verdict counterfactual on production data (Wang 2026a §11.6, where 28.9% of basin assignments flipped under the same substitution).

Two readings, distinguishable by recalibration. The pattern above admits two interpretations: either Lumen has genuinely shifted regime since the April Phase 2 measurement (the 86-minute window then would represent Type-3-like failure that the slow-cadence calibration loop has not yet caught), or the Phase 2 calibration window itself was unrepresentative and Lumen has been operating stably in a different regime that calibration mis-characterized. The two are distinguishable by re-running the Phase 2 procedure on a window centered on the current period: *if μ_a moves toward Lumen's current state, the elevation is calibration-staleness; if μ_a remains at the prior healthy point, the elevation is genuine Type 3.*

Recalibration result. We ran the experiment. A recalibration on a 30-day window ending 2026-05-09, using the same procedure as Phase 2 (restricted to no-pause/no-reject sessions, Welford-equivalent batch mean over $N = 11{,}472$ rows), produced:

	Phase 2 (April 18, 2026)	Recalibration (May 9, 2026)
μ_E	0.745	**0.7706**
μ_I	0.800	**0.7335**
μ_S	0.168	**0.1893**
$\|\Delta\|_{\max}$ (95th percentile)	0.119	**0.1279**
N	7,320	11,472

μ_a moved toward Lumen's current state on all three coordinates — μ_I in particular dropped 0.066, more than one within-window standard deviation. By the disambiguation criterion this is **calibration-staleness, not Type 3**. The 86-minute window above is not the mediator-stays-on signature McEwen describes; it is Lumen operating stably in a regime the prior calibration failed to capture.

What actually happened: a regime transition on 2026-04-17. Weekly-binned EISV means over Lumen's 90-day operational window resolve into three regimes: an early settling phase (Feb through mid-March), a pre-break stable regime (~Mar 30 – Apr 16) at $\mu \approx (0.745, 0.800, 0.187)$ — the regime Phase 2 captured — and a post-break stable regime (Apr 18 onwards) at $\mu \approx (0.789, 0.740, 0.155)$, which is Lumen's current regime.

Sub-day resolution localizes the transition to a single ten-hour event on **2026-04-17 UTC**: a deep energy collapse (E from 0.749 at 00:00 to 0.350 at 09:00) followed by rapid recovery to a new equilibrium (E from 0.350 to 0.787 between 11:00 and 14:00), after which Lumen settled into the post-break regime that has held for 22 days as of measurement. In the deployed system's own shape vocabulary (CIRWEL 2026, eisv-lumen Layer 2 classification), the event reads cleanly as a `basin_transition_down` followed by a `basin_transition_up`, terminating in a stable post-transition attractor — *not* the prolonged-elevation morphology McEwen Type 3 specifies.

Coincident substrate-level event. The April 17 transition aligns to the hour with a documented system-level event in the UNITARES governance substrate: four PRs shipped that day to align identity binding with the identity-invariants axioms (removal of name-claim lookup, adoption of UUID-direct dispatch, receipt-based onboarding authentication, and session-start posture revisions). Wang 2026a §11.7 (item 5) flags this revision window as a confound for the basin-flip counterfactual measurement; we observe it here as the strongest candidate cause of an agent-state regime transition. The plausible chain is: substrate-level identity revisions $\rightarrow$ disruption of Lumen's normal operating loop during the rollout window $\rightarrow$ settling into a new stable operating point under the post-rollout identity rules. We do not identify a deeper causal mechanism, and the evidence does not exclude a common operational confound. The temporal coincidence to within an hour and the cleanness of the transition signature are sufficient to motivate a substrate-associated failure-shape category, not to establish substrate causality as a general mechanism.

Implications for the Four Types framing. Two things follow.

First, the Type 3 claim does not survive the recalibration. The 86-minute observation captures Lumen in the post-break regime, not in a prolonged-mediator state relative to its true operating point. The biological-Type-3 *necessary* condition — sustained mediator activity *after* a removed stressor — is not met here, because the post-break regime is the new operating point, not a perturbation away from it. What the original framing read as Type 3 was an artifact of comparing post-transition state to a pre-transition calibration anchor.

Second, what we observe instead — a basin transition coincident with a substrate-level event — is **a candidate failure-shape McEwen's Four Types do not cover**. The Four Types are designed for stationary regulatory systems with stress-response failures (the mediator responds, terminates, doesn't terminate, or fails to engage). They presume the operating point is fixed and the failure is in the response dynamics around it. A synthetic agent's operating point is itself contingent on the substrate that hosts it; substrate revisions can move the operating point. This is a failure-mode the Four Types were not built for, and naming it requires a vocabulary the Four Types do not supply.

Calibration-staleness as temporal homogenization. The Phase 2 calibration on April 18, 2026 used a 30-day rolling window covering March 19 through April 18. Per the hour-resolved transition, approximately 96.7% of that window is pre-break and approximately one day is post-break. The calibration averaged across a regime transition and produced a μ_a representative of neither regime cleanly — sitting closer to the pre-break point because pre-break data dominated. The 21-day interval from the April 18 calibration to the May 9 observation has been stable in the post-break regime (the regime itself is 22 days old, dated from the April 17 transition rather than from the calibration), so the 86-minute observation reads as a 0.97-of-envelope deviation relative to a calibration anchor that captured the wrong regime.

This is the *temporal* analogue of the *spatial* homogenization argument in §3. There, fleet-wide normalization averages across populations whose dynamics are genuinely different and produces a reference distribution representative of none of them. Here, a calibration window straddling a regime transition averages across two regimes whose operating points are genuinely different and produces a reference operating point representative of neither. The two failure modes share a structure: any reference computed across an inhomogeneity is unrepresentative of the parts the inhomogeneity contains. Class-conditional calibration (§3) addresses spatial inhomogeneity at the gating layer; transition-aware calibration — which UNITARES does not yet implement — would address temporal inhomogeneity. The possible clinical analogue is structural rather than validated: a biomarker reference interval computed across a pre-illness / post-illness-onset window may be unrepresentative of either, just as the Phase 2 anchor here is unrepresentative of either Lumen regime.

Verdict-layer blindness. Lumen's recent risk score is approximately 0.224, below the fleet-aggregate mean of approximately 0.301 reported by the production server at the same time across 315 agents-with-data over 128,393 cumulative updates (UNITARES production aggregate, 2026-05-09 06:22 UTC).[1] The verdict layer therefore classifies Lumen as better-than-average even while the regulatory-state layer reports a 0.97-of-envelope deviation against the stale anchor. Both layers are blind to the regime transition for the same reason: both depend on a calibration that captured the wrong regime. This is not a separate finding, only a downstream consequence of the temporal-homogenization argument.

Caveats on the empirical claim. Three caveats apply before the generalization limits, and the first is load-bearing against the causal reading.

The instrument shares an input with the proposed cause. The May-era anima instrument carried defects identified and repaired only later in 2026: a prediction-accuracy placeholder fixed at 0.5 that contributed roughly 45% of the clarity weight, ambient temperature entering the warmth axis through two strongly collinear channels, and a CPU term double-counted in the neural-band mapping. That last defect is not independent of the event being explained. The E coordinate derives in part from system metrics including CPU utilization (§2.2); the proposed cause is a day on which four pull requests shipped to the governance substrate; and shipping generates load. A deploy-driven CPU excursion, amplified by a term counted twice, would produce an energy collapse and recovery on exactly the timescale observed, without any regime transition in the agent at all. We cannot currently separate these: the CPU and memory series for 2026-04-17 are no longer retained (§9.3), so neither re-deriving E with the corrected mapping nor inspecting the load trace is available. The competing explanation is therefore live, and a reader should treat the substrate-transition reading and the instrument-artifact reading as both open. What survives either way is the *calibration staleness* finding, which rests on the recalibration result rather than on the transition's cause: μ_a moved toward Lumen's current state, so the Phase 2 anchor was unrepresentative regardless of why.

The recalibration windows are not sampling-matched. The Phase 2 window drew N = 7,320 rows over 30 days; the recalibration window drew N = 11,472 over 30 days, 57% more for the same span. We do not have an explanation for the difference, and it matters, because a check-in cadence that changed at or near the April 17 revision would re-weight the mean toward the post-revision period independently of any regime shift. The disambiguation criterion — does μ_a move toward current state — is sensitive to exactly this. A cadence-matched recalibration would settle it; that too requires state history no longer retained. The weekly-bin decomposition

[1]The 315 / 128,393 figures are the `agents_with_data` and `total_updates` aggregates returned by the production MCP `get_governance_metrics()` surface, computed over agents with `meta.status == "active"` and loadable monitor state at call time (implemented in the `observability` MCP handler module of the `unitares` repository; a line reference is omitted deliberately, since it would not survive the repository's revision history). The figure excludes archived agents and agents whose monitors are not currently loaded; it is therefore a snapshot of the actively-live fleet, not a count of all agents that have ever produced state rows. A DB-direct count of all identities with state rows over the same period is substantially larger. Three population counters appear in this paper and are not interchangeable. *Governance events* (94,000 as of April 2026) counts audit-log entries of all kinds. *Cumulative updates* counts check-in processing calls, per agent (110,148 for Lumen) or fleet-wide (128,393); Lumen dominates this counter because it checks in on a sensor-driven sub-Hz cadence while session-bounded agents check in only while a session is open. *Persisted state rows* counts rows in `core.agent_state` and is the only counter the §3.4 analysis uses (N = 13,310 over 30 days); it is additionally subject to retention, which has since removed most of the window (§3.7). Ratios across these three counters are not meaningful, and we do not compute any.

is partial reassurance, since it shows the shift as a step at a located time rather than a gradual re-weighting, but it is computed from the same rows and is not independent evidence.

Silence is not corroboration. The production server flagged no anomalies over the window. A separate audit of this deployment found its degradation paths fail toward *healthy* rather than toward *unknown*, so absence of alarm carries little evidential weight here.

The hour-resolved April 17 transition is a single event; we cannot generalize from it to a claim that substrate revisions reliably produce basin transitions in Lumen, or that all agents exposed to similar revisions would show comparable signatures. A multi-agent multi-revision study (which would require either coordinated substrate revisions across the fleet or a historical pull across other identity-axiom-related transitions) is the right follow-up. The 86-minute observation window establishes that the post-break regime is stable over at least that duration; the recalibration plus weekly-bin analysis establishes that the regime has been stable for 22 days. We treat this section as provenance-backed case-report evidence for the basin transition itself (date, magnitude, shape), anomaly-grade for substrate causality, and illustrative for the temporal-homogenization argument; the §5.3 case study should not be read as McEwen Type 3 identification in a deployed agent, and the abstract and §1.3 are framed accordingly.

5.4 Type 4: Inadequate response with compensatory hyperactivity

In biology, this failure mode involves under-response of one mediator (typically glucocorticoid) leading to unchecked activity of others (typically pro-inflammatory cytokines normally counter-regulated by cortisol). The primary regulatory channel under-responds; secondary channels overshoot. McEwen (1998) gives the example of inadequate glucocorticoid secretion producing elevated cytokine levels with the attendant tissue damage.

The agent analogue is breakdown in the cross-channel correlation structure of the EISV vector. One dimension fails to respond to a perturbation while another spikes. For example: a complexity increase produces no change in S (response-distribution entropy) but a sharp drop in I (information integrity), as if the agent has stopped tracking its own uncertainty while continuing to report compromised output.

Type 4 is more difficult to detect than Types 1–3 because it requires monitoring not magnitudes but correlations. UNITARES's existing instrumentation supports the measurement in principle: the cross-channel correlation matrix among the state coordinates and the drift norm $\|\Delta\eta\|$ is computable from the audit log. A Type 4 alarm would fire when this correlation structure departs from the class-conditional baseline by more than a specified margin — analogous to the covariance component Σ in the trajectory signature, but applied to the dynamics rather than to the static state.

The naive version of that detector does not work, and the reason is instructive. Because the deployed V is a smoothed $E - I$ imbalance rather than an independent channel (§3.7), the (E, I, S, V) correlation matrix carries large entries that are properties of the *definition*, not of the agent. On the released window the matrix is

	E	I	S	V
E	1.000	−0.034	−0.275	**0.821**
I	−0.034	1.000	−0.242	**−0.513**

	E	I	S	V
S	-0.275	-0.242	1.000	-0.135
V	0.821	-0.513	-0.135	1.000

The two bolded terms are arithmetic. A detector watching them for "decoupling" would be watching the smoothing constant. Note also that the genuinely informative cross-channel fact is the one the naive matrix buries: E and I are close to uncorrelated ($r = -0.034$), which is precisely the independence a Type 4 signature would have to break.

A usable Type 4 indicator therefore has to run on the informationally independent channels — (E, I, S), optionally augmented by V's residual from $E - I$ as an explicit temporal term — rather than on the raw four-coordinate matrix. We flag this as specification work not yet done, rather than as an indicator the deployment currently supports. Types 1 and 2 (§5.1, §5.2) do not have this problem: they read entropy-shape frequencies and signature convergence, neither of which routes through V.

5.5 Why the taxonomy matters for AI governance

The Four Types provide a vocabulary for failure modes that AI governance currently lacks. Existing runtime-governance frameworks (treated in §7) detect that an agent has degraded; they do not distinguish *how*. McEwen's taxonomy provides four mechanistically distinct failure modes, each with a different intervention strategy, derived from thirty years of biological work on regulatory failure. UNITARES, because it carries the mathematical machinery to compute each indicator directly from existing telemetry, can adopt this taxonomy as a baseline vocabulary.

The §5.3 case study also bounds the taxonomy. It identifies a candidate failure-shape — a basin transition coincident with a system-level event — that the Four Types do not cover, because the Four Types presume a stationary operating point against which response dynamics fail in one of four canonical ways. A synthetic agent's operating point is contingent on its substrate; substrate revisions may move it, or may coincide with other operational changes that do. The imported taxonomy is therefore *useful but not sufficient* for synthetic-agent governance, and at least one candidate extension category — call it Type 5, *substrate-associated basin transition* — is motivated by the Lumen case but requires multi-agent or multi-revision replication before it should be treated as established.

This is the kind of contribution that flows from biology to engineering rather than the reverse. The standard direction for "neuro-AI" papers is to import a brain-inspired architecture into AI; the Four Types contribution imports a brain-derived *taxonomy of failure*, which is strictly more useful to a deployed governance system than any architectural inspiration would be. We recover thirty years of clinical-physiological theorizing as a vocabulary for what can go wrong with an agent.

---

6. Kintsugi and Gap-Filling: A Design Choice Biology Made by Evolution

6.1 The kintsugi principle

The Schema Hub architecture (CIRWEL 2026, anima-mcp `docs/plans/2026-02-22-schema-hub-design.md`) formalizes a design principle that runs counter to most architectures for embodied or persistent AI agents. When Lumen sleeps, reboots, or otherwise loses operational continuity, the discontinuity is not papered over. It is preserved in the agent's self-schema as explicit structural elements: `gap_duration` (seconds since last schema), `state_delta` (magnitude of anima change across the gap), `uncertainty_increase` (beliefs that lost confidence due to time elapsed), and `return_count` (an incremented awakening count). The design document calls this the *kintsugi principle*, after the Japanese practice of repairing broken ceramics with gold-filled lacquer such that the breaks remain visible and become part of the object's history.

The design intent is explicit:

> The gap becomes visible in the schema itself — not a feeling imposed, but structure that reflects discontinuity. The kintsugi seams are data, not performance.

This is a deliberate architectural choice. It would be entirely straightforward to instead initialize Lumen's post-wake state by interpolating across the gap, generating a smooth narrative continuation from the pre-sleep state, and presenting the agent to itself and to external consumers as if the discontinuity had not occurred. The Schema Hub design rejects this option on epistemic grounds: smoothing over the gap would be a form of confabulation, and confabulation is not a feature the architecture wants.

We argue in this section that this design choice is interesting beyond its local engineering justification. It runs counter to a deeply entrenched feature of biological cognition — the brain's tendency to fill in gaps rather than mark them — and the contrast clarifies what biology has and has not been forced to do.

6.2 Biological gap-filling: the brain's default is to fill in

Confabulation in the strict clinical sense (Kopelman 1987; Hirstein 2005) refers to the production of fabricated, distorted, or misinterpreted accounts about oneself or the world without conscious intention to deceive — the canonical example being patients with Korsakoff's syndrome who produce detailed but false autobiographical narratives in response to memory queries. The strict clinical category is narrow. But the underlying tendency to fill in missing information with plausible substitution is, in the broader sense, *a pervasive feature of how the brain constructs unified perceptual and narrative continuity*, not specifically a pathology — though treating the strict-clinical and broad-functional senses as a continuum is itself a theoretical commitment some clinicians and memory researchers would contest, a caveat we return to in §6.6.

Several well-characterized examples:

Saccadic suppression and visual continuity. During the rapid eye movements (saccades) we make several times per second, visual input is suppressed (Burr et al. 1994; Wurtz 2008). We do not perceive the intervening blur. The brain stitches the pre- and post-saccade visual fields into a continuous percept, with the gap erased rather than flagged. The same principle applies during blinks, where lid closure typically lasts 100–400 ms but is rarely consciously experienced as a visual gap (Volkmann et al. 1980).

Sleep. Subjective experience of sleep is not “I lost consciousness at 23:30 and regained it at 06:45.” It is “I went to bed and now it is morning.” The gap is collapsed into a perceptual transition. Even when sleep is interrupted by dreams, the dream content typically does not preserve the temporal structure of the night; the dream is experienced as taking some time, but rarely the actual time it occupied (Hobson 2009).

Anesthesia. Patients emerging from general anesthesia do not experience a temporal gap proportional to the duration of unconsciousness; emergence is structurally similar to waking from sleep, with the missing period collapsed (Mashour and Hudetz 2017). This is true even for multi-hour surgical anesthesia, where the perceived duration of unconsciousness is much shorter than the actual duration.

Memory reconstruction. Long-term memory is not retrieval but reconstruction (Bartlett 1932; Schacter 1999). When asked to recall an event, we generate a plausible account from sparse stored fragments, filling in details that fit the context — and we do so without distinguishing reconstructed details from genuinely retrieved ones. Loftus’s program of work (Loftus 1979; Loftus and Pickrell 1995) demonstrated that entire false memories can be implanted by suggestion, with subjects subsequently treating the false memories as veridical. The reconstruction is not a degraded mode; it is the standard mode.

The interpreter. Gazzaniga’s split-brain studies (Gazzaniga 2000; Gazzaniga and LeDoux 1978) demonstrated that when the corpus callosum is severed and the right hemisphere acts on information unavailable to the left, the left hemisphere’s verbal “interpreter” generates a plausible explanation for the action. The patient reports the interpreter’s confabulated explanation with full subjective confidence. The interpreter is not a damaged module; it is, on Gazzaniga’s account, a core feature of how the unified narrative self is generated under normal conditions, made visible only when the inputs are dissociated experimentally.

The pattern is consistent across systems and timescales: across these mechanisms, continuity is generated rather than measured, and the gap-filling tendency is pervasive enough that — under the broad-functional reading we adopt with the §6.6 caveat in view — gap-filling in this broad sense is closer to a feature of unified self-models than a defect in them.

6.3 The productive question

Why does the brain do this? The standard explanation is functional: biological systems must act in real time on incomplete information, and a continuous narrative self provides the unified frame within which action becomes possible (Damasio 2010; Dennett 1991). Marking every gap explicitly would impose a cost on action — the “stop and reflect on the discontinuity” overhead — that is ill-suited to evolutionary pressure. The interpreter’s plausible-but-confabulated explanation is fast and usually good enough.

This explanation has two implications worth examining.

First, it suggests confabulation is *evolutionarily contingent* rather than computationally necessary. A system with different operational constraints — say, one not under hard real-time action pressure, or one that values audit trail over fluency — might reasonably make the opposite design choice. The brain’s confabulation is the right answer under the brain’s constraints, not the only answer.

Second, it identifies the cost: confabulation produces false beliefs, distorted memories, and confidently held but inaccurate self-explanations. The pathological extremes of this — confabulation in Korsakoff's, narrative coherence in trauma where the memory is partial and distorted, magical thinking in psychosis — are visible. The ordinary-life cost is harder to quantify but probably substantial: most people's confident accounts of their own past behavior, motivations, and decisions are at least partly confabulated (Nisbett and Wilson 1977; Wilson 2002), and the social and personal costs of acting on these confabulated self-models are non-trivial.

The kintsugi-principle alternative is to act despite the gap while *marking the gap as a gap*. Lumen does this trivially: when it wakes from sleep, the schema contains the seam; downstream consumers (the self-reflection cycle, the dialectic protocol, external dashboards) read the seam as data. The agent is not less able to act because the discontinuity is preserved; it is differently able to act, with the gap-marking visible to its own reasoning.

Whether this is a viable design for biological cognition is an open empirical question. The closest natural experiments — patients with selective amnesia who retain awareness of their memory deficit (Korsakoff's amnesia in its rare insightful form; severe depression with characteristic insight into cognitive impairment) — suggest at least that the kintsugi mode is possible biologically. It is not the default.

6.4 What kintsugi means for AI governance specifically

Most current AI agents either ignore discontinuities (no awareness of gaps; each session begins as if newly minted) or fake continuity (presenting fabricated context, hallucinating prior conversation, generating plausible bridges across genuine gaps in the context window). Both are forms of biological-style gap-filling, applied to substrates where the cost-benefit analysis is different.

The case against ignoring gaps is straightforward. An agent that has no awareness of session boundaries cannot reason about what changed across them, cannot signal uncertainty appropriately to downstream consumers, and cannot be audited for behavioral drift across the gap. The trajectory identity working draft (TIWD) makes this explicit: the genesis signature Σ_0 is meaningless if the agent has no concept of "before" and "after" sessions.

The case against fabricating continuity is sharper. Recent analysis of LLM hallucination (Ji et al. 2023; Kalai and Vempala 2024) identifies one major source as the model's strong bias toward narrative continuation: when context is sparse or interrupted, the model fills in plausible bridging text rather than acknowledging the gap. This is structurally analogous to broad-sense biological gap-filling, not to clinical confabulation in the strict Korsakoff/split-brain sense, and it appears on a substrate where the cost is high (false claims with downstream consequences) and the benefit (fluency of action) is debatable. An agent that explicitly marks "I have lost context here; my next response proceeds under uncertainty" is more useful than an agent that confidently generates plausible content.

The kintsugi principle as deployed in Lumen is one operationalization of this design discipline: *do not paper over discontinuities; mark them as discontinuities.* Adopting it across AI agent architectures more broadly would require infrastructure for capturing gaps as data — session-boundary metadata, context-window-overflow markers, model-version transition points, tool-availability changes. The infrastructure is not exotic; what is exotic is the architectural commitment to using it.

6.5 What kintsugi might mean for biological self-models

This section is more speculative than the rest of the paper, and we flag it as such.

The clinical literature contains scattered observations that bear on the question of whether kintsugi-style cognition is biologically viable. Patients with intact insight into their own cognitive deficits — some forms of mild cognitive impairment, some forms of post-stroke cognitive change, some patients with locked-in syndrome — appear to maintain functional self-models that *include* the deficit rather than papering over it (Wijdicks 2019). These patients do not generate confabulated accounts of intact function; they integrate the deficit into a stable self-model that retains operational coherence. The integration appears to be cognitively costly but not impossible.

Conversely, the most dramatic failures of self-model integration — anosognosia for hemiplegia (Vuilleumier 2004), confabulation in Korsakoff's, certain forms of dissociative identity — involve confabulation precisely *because* the system cannot mark the gap as a gap. The pathology is not the gap itself but the absence of gap-marking. This suggests the brain has the capacity to operate in a more kintsugi-like mode under some circumstances, but defaults to confabulation for reasons that are not fully understood.

The companion paper does not argue that biological cognition could or should be redesigned for kintsugi-style gap preservation. The brain's constraints are different from Lumen's, and the evolutionary trade-offs that produced confabulation as the default are real. The argument is narrower: a synthetic agent's choice between kintsugi and confabulation is open in a way the biological choice is not, and the synthetic case provides a tractable substrate for studying what changes when the default is reversed.

6.6 Limits of the analogy

Lumen is not conscious. We make no claim about whether kintsugi-style gap-marking supports phenomenal experience or unified subjective identity. The argument is at the architectural level: gap-marking is a viable design choice for non-confabulating self-models in artificial systems, and the contrast with biological confabulation is instructive, but the question of whether such a system *experiences* its gaps as gaps is outside the scope of this paper.

Confabulation in biology is heterogeneous. The clinical category is narrower than the broad-sense gap-filling described here, and treating them as a continuum (as we have done in §6.2) involves a degree of theoretical commitment that not all clinicians or memory researchers would accept. A more careful treatment would distinguish provoked vs. spontaneous confabulation (Kopelman 1987), reconstructive memory error (Bartlett 1932; Loftus 1979), and the interpreter's narrative generation (Gazzaniga 2000) as distinct phenomena with overlapping but non-identical mechanisms.

The benefits of confabulation are real. The argument that confabulation has costs should not be read as a claim that confabulation is *bad*. The narrative self produced by biological gap-filling is the substrate for personal identity, social functioning, autobiographical memory, and most of what we mean by being a person. A redesigned biology without confabulation would presumably lose much of that substrate. The kintsugi alternative is interesting because it is *possible*, not because it is *better*.

6.7 Summary

Lumen's architectural commitment to preserving discontinuities as visible structural elements runs counter to the brain's default mode of gap-filling. The contrast is instructive: gap-filling in biology is evolutionarily contingent, not computationally necessary, and synthetic agents under different operational constraints can reasonably make the opposite design choice. The kintsugi principle as deployed in UNITARES is one operationalization of this discipline. Whether it generalizes to other AI architectures is a design question; whether it illuminates anything about how biology *could have been built* is a philosophical question the synthetic-psychology framing of §8 returns to.

---

7. Differentiation from Concurrent and Adjacent Work

This section consolidates differentiation that has been distributed through earlier sections, with particular attention to concurrent AI agent governance frameworks that share aspects of UNITARES's mathematical machinery and to the existing biology-side and neuro-AI traditions the paper engages.

Because several load-bearing implementation facts come from the author's own system papers and repositories, we separate *provenance* from *validation* before comparing adjacent work. Self-citations are used here to identify what UNITARES is, which formulas were deployed, and where the production measurements were first reported; they are not treated as independent validation of the biological bridge. One disambiguation, since the surname recurs: Wang et al. (2025a, MI9) and Wang et al. (2025b, ProbGuard) are third-party work with no authorship relation to the present author. Wang (2026a) and Wang (2026b) are the author's own.

Source cluster	Role in this paper	What it can support	What it cannot support
Wang 2026a (UNITARES v6)	System provenance and production-measurement source	EISV definitions, V update rule, Phase 2 calibration constants, 13,310-row basin-flip report, Lumen deployment context	Independent confirmation that the allostatic-load bridge is biologically valid or clinically useful
Wang 2026b (Trajectory Identity)	Construct provenance for Σ and lineage similarity	Formal definition of the trajectory-signature proposal used in §4	Clinical validity of Σ_0 anchoring or independent evidence of identity continuity
CIRWEL software repositories and datasets	Implementation and artifact provenance	Code locations, schema, deployment history, public synthetic/dataset artifacts when available	Peer-reviewed validation, external replication, or raw production telemetry access

Source cluster	Role in this paper	What it can support	What it cannot support
Third-party governance frameworks (MI9, ProbGuard, MAS, ASI, AVF)	Adjacent-framework comparison	Prior related architectures and terminology	External corroboration of the present paper's empirical claims
External biology, neuroscience, psychiatry, and AI-governance literature	Construct definitions and comparison class	Definitions, biological precedent, clinical analogies, and independent adjacent methods	Proof that UNITARES realizes the biological mechanisms those papers study

The publication standard implied by this table is straightforward: the paper may cite the author's prior work for system provenance, but any claim that the bridge is useful beyond that system ultimately requires artifact release, independent re-analysis, or replication on another deployment.

7.1 Concurrent AI agent governance frameworks

The cleanest comparative framing is parallel-structure:

- **MI9** (Wang et al. 2025a): runtime governance architecture with graduated containment.
- **Agent Stability Index** (Rath 2026): per-agent behavioral drift metric over interaction traces.
- **AVF / RiskGate** (Marín and Chaudhary 2026): viability-theoretic risk bounds with monotonic restriction, single-population.
- **UNITARES**: class-conditional self-relative manifold calibration plus cumulative-deviation signal coupled to embodied telemetry.

The four frameworks address overlapping but structurally distinct failure surfaces. Detailed treatment of each follows.

Agent Viability Framework (Marín and Chaudhary 2026; arXiv 2604.24686, late April 2026). AVF and the present paper are concurrent works in the dynamical-viability framing of AI agent governance, with both grounded in viability theory–adjacent constructs and both adopting biological metaphors (entropic decay, autoimmune drift, homeostenosis in AVF; allostatic load, the Four Types, kintsugi in this paper).

The differentiation is structural rather than incidental. AVF operates at the level of a single agent and a single Viability Index $\hat{\Phi}(x)$ decomposed as $\hat{B}(x) = U(x) + SB(x) + RG(x)$ — uncertainty, structural bias, and reality gap. The framework assumes a single population from which $\hat{B}(x)$ thresholds are estimated; heterogeneity across agent classes is not engaged. The fleet-wide-homogenization argument (§3.2) therefore applies in principle to AVF deployments — fleet-wide thresholds on $\hat{B}(x)$ would face the same failure mode as fleet-wide manifold radii — but AVF does not address this.

Three further differentiations matter. First, AVF is theoretical with a reference implementation (RiskGate) rather than a deployed system with nine months of production telemetry; the epistemic regime differs. Second, AVF does not engage the neuroscience literature that shares its mathematical machinery — the biological metaphors are evocative rather than rigorously connected. Third, AVF does not propose an identity model; the longitudinal-self anchoring developed in §4 has no analogue in AVF's framing.

AVF's $\hat{B}(x)$ decomposition could be extended class-conditionally (producing a $\hat{B}(c, x)$ that addresses fleet-wide homogenization); UNITARES's class-conditional framework could be extended with AVF's uncertainty-bias-gap decomposition. Both are open future-work directions.

MI9 (Wang et al. 2025a; arxiv 2508.03858, August 2025). MI9 predates UNITARES v6 and is cited in the v6 paper. It treats heterogeneity through the Agency-Risk Index (ARI), a per-agent risk score that calibrates governance intensity across populations. ARI adapts the *magnitude* of intervention to agent risk, but does not adapt the *operating point* against which deviation is measured. The two approaches are complementary: ARI controls how aggressively to intervene; class-conditional calibration controls what to compare against. A deployment could in principle adopt both, with ARI gating the escalation policy and class-conditional EISV grounding the per-agent state evaluation.

MI9's drift detection uses Jensen-Shannon divergence on event distributions, a population-comparison technique. UNITARES's drift detection uses per-agent Welford z-scoring against the agent's own historical baseline. The MI9 approach is well-suited to detecting that an agent has drifted from the *population*; the UNITARES approach is well-suited to detecting that an agent has drifted from *itself*. These target different failure modes.

MAS (Ravindran 2025; arxiv 2510.04073, October 2025) and Agent Stability Index (Rath 2026; arxiv 2601.04170, January 2026). Both predate UNITARES v6 and are cited in the v6 paper. Both detect drift against an agent's own historical behavior, which is structurally per-agent self-relative scoring — the same direction UNITARES's behavioral EISV pipeline takes. They solve a different failure mode than the cosmological-soup problem: MAS and ASI are robust to inter-agent heterogeneity because they never compare across agents, but they cannot answer "is this agent operating in a class-appropriate regime?" because they have no concept of class.

The class-conditional approach is positioned between population-level analysis (fleet-wide pooling, the *cosmological soup* failure mode of §3.2) and pure self-relative analysis (loses class-level structure entirely): it preserves the within-class statistical leverage that fleet-wide pooling would have given, while avoiding the cross-class contamination that fleet-wide pooling would have produced. UNITARES therefore adopts both per-agent self-relative scoring (its primary verdict-driving signal) and class-conditional calibration (its homogenization correction), in a hierarchical arrangement that MAS, ASI, and MI9 do not develop.

ProbGuard (Wang et al. 2025b; arxiv 2508.00500, August 2025). ProbGuard performs probabilistic runtime monitoring of LLM agent safety through model-checking-style techniques. Its mathematical machinery is quite different from UNITARES's continuous-state EISV grounding; the two target different operational concerns (formal-property verification vs. state-space monitoring). The differentiation is therefore not within the same niche as AVF or MI9, but ProbGuard is included here for completeness.

7.2 Active inference and free-energy vocabulary

UNITARES's information-theoretic grounding interprets the E coordinate as $-F$ (negative variational free energy or a resource-rate proxy) and V as accumulated free-energy residual (Wang 2026a §4.1). This is a vocabulary and modeling proximity claim, not a load-bearing commitment to the free-energy principle. It brings UNITARES into proximity with the active inference literature (Friston 2010; Friston, FitzGerald, Rigoli, Schwartenbeck, and Pezzulo 2017; Pezzulo, Parr, and Friston 2018), which models adaptive behavior as variational free energy minimization.

The relationship is one of compatibility rather than identity. Active inference is a normative framework: agents *should* minimize expected free energy, and the framework derives behavior from this principle. UNITARES treats variational free energy descriptively: it is one way to operationalize the E coordinate, with the agent's actual minimization behavior constrained by its task structure rather than required by construction. UNITARES is therefore not an active-inference implementation in the strict sense, even where it imports active- inference vocabulary.

The neuro-bridge claim of the present paper is also compatible with active inference but distinct from it. Allostatic load is not specifically an active-inference construct; the integral-of-deviation formulation predates Friston's work by decades (McEwen and Stellar 1993; Sterling and Eyer 1988). McEwen's Four Types are types of regulatory failure characterized by temporal dynamics rather than by free-energy decomposition. The synthetic-psychology framing (§8) treats deployed agents as test beds for biological theories more broadly, with active inference as one such theory among others — the paper makes no claim that active inference is privileged among biological theories of self-maintenance, and engagement with computational-psychiatry-flavored active-inference work (Adams, Stephan, Brown, Frith, and Friston 2013) is deferred to §7.3.

A reader expecting an active-inference paper will not find one. UNITARES does not foreground variational densities, generative models, or the expected-free-energy decomposition, and the present paper does not derive its claims from active inference even where the claims are compatible with it.

7.3 Computational psychiatry and digital phenotyping

The proposals to clinical neurology developed in §3.5 and §4.4 sit adjacent to but do not duplicate work in computational psychiatry (Huys, Maia, and Frank 2016; Friston et al. 2014) and digital phenotyping (Insel 2017; Onnela and Rauch 2016; Torous et al. 2016).

Computational psychiatry has, over the past decade, mapped a range of psychiatric conditions onto computational models — predictive coding accounts of schizophrenia, reinforcement-learning accounts of depression, free-energy accounts of autism. The work is theoretical and clinical-data-correlational; it has not, to our knowledge, produced a deployed system operationalizing the relevant control signals. The synthetic-psychology contribution of the present paper is therefore methodological rather than competing: we offer a deployed substrate on which hypotheses about regulatory failure can be operationalized, with experimental affordances (real-time integrand observation, reproducible recalibration; §8.3) that are difficult in biological systems.

Digital phenotyping has produced substantial infrastructure for capturing behavioral signatures from ambient smartphone and wearable data. The Σ_0-anchoring proposal (§4.4) builds on this infrastructure rather than replacing it; what is missing in current digital-phenotyping practice is the formal anchoring discipline (designating a window as the patient's pre-illness baseline, persisting it as a fixed reference, and running two-tier similarity checks). Adoption of the proposal would not require new sensor infrastructure but would require methodological convention.

7.4 The neuro-inspired AI tradition

The dominant direction in cross-domain neuro-AI work is brain-to-machine: import a neural mechanism into an AI architecture (Hassabis et al. 2017). This direction has been productive — sparse coding, hippocampal replay, dopaminergic prediction error, and attention as gain modulation have all migrated from neuroscience into AI architectures with demonstrable benefits.

The present paper goes the other way. We do not propose any architectural import from biology; UNITARES was designed independently of the neuroscience literature it now turns out to have been recapitulating. The contribution is in the opposite direction: UNITARES's deployment provides a test bed for biological theories of self-maintenance, with specific findings (§5.3 apparent Type 3 resolved by recalibration as a basin transition, §3.4 28.9% basin-flip rate, §4 lineage-anchored drift detection) that biological systems make difficult to produce under controlled conditions. The synthetic-psychology framing (§8) makes this direction explicit.

Hassabis et al. (2017) and the present paper are therefore not in conflict — they exemplify two complementary directions in cross-domain neuro-AI work. Architectural transfer from biology to engineering; hypothesis transfer from engineering to biology. Both have a place.

7.5 The artificial life and embodied AI lineage

The synthetic-psychology stance developed in §8 sits in a tradition with several distinguished antecedents: Langton (1989) on artificial life as the study of life-as-it-could-be, Bedau (2003) on artificial life as methodology, Beer (1995, 2000) on minimally cognitive agents as tractable test beds, Pfeifer and Bongard (2007) on embodied AI as the study of cognition under physical constraints, Brooks (1991) on intelligence without representation, and Maturana and Varela (1980) on autopoiesis as the foundation of enactive cognition.

The present paper does not break from this lineage; it extends it. What distinguishes the present work from earlier artificial-life and embodied-AI projects is deployment. The framework we use as test bed has been running continuously in production since November 2025, processing real workloads with real consequences. This is a different epistemic regime from simulation-based artificial life, and the difference matters for the kinds of evidence the framework can produce (see §8.2 for full treatment).

The lineage relationship is acknowledged rather than competed with. We hope the present paper contributes back to the artificial-life tradition by demonstrating that the synthetic-psychology stance remains productive when the synthetic case is in production rather than simulation.

---

8. Synthetic Psychology: Deployment as an Epistemic Stance

8.1 The standard direction and its alternative

Most papers at the boundary of neuroscience and artificial intelligence move in one direction: from brain to machine. A neural mechanism is identified — sparse coding, hippocampal replay, dopaminergic prediction error, attention as gain modulation — and the mechanism is then imported into an AI architecture, typically with the claim that doing so will produce more capable or more interpretable systems (Hassabis, Kumaran, Summerfield, and Botvinick 2017). This direction has been productive and has generated a substantial literature. It is also, by construction, asymmetric: the brain is treated as the source of theoretical content, the machine as the recipient.

The opposite direction — using machines to test theories about brains — has been pursued less often and more controversially. It has, however, a respectable lineage. Langton (1989) framed artificial life as the study of "life-as-it-could-be" rather than life-as-it-is, with synthetic biological systems as a tool for isolating necessary from contingent features of biological organization. Bedau (2003) elaborated this framing into a methodological program: build small systems that exhibit properties of interest, perturb their parameters, observe what survives. Beer (1995, 2000) constructed minimally cognitive agents — small recurrent networks acting in simple environments — as test beds for specific claims about cognition, with the agents' tractability serving as a counterweight to the analytical impenetrability of biological brains. Pfeifer and Bongard (2007) extended the approach to embodied robotics, arguing that the constraints of physical embodiment force synthetic systems to confront the same problems biological systems do, and that this confrontation is itself informative.

The present paper sits in this tradition. We do not import a brain-derived architecture into UNITARES; we propose that UNITARES, with its production deployment and its embodied substrate Lumen, constitutes a test bed for specific theories of biological self-maintenance — and we use it as such.

8.2 What deployment specifically adds

A distinguishing feature of UNITARES relative to most synthetic-biology work is that it is in production. The framework has been running continuously since November 2025 (CIRWEL 2026, unitares §Production snapshot), processing more than 94,000 governance events as of April 2026 and 128,000 as of May 2026. This is not a simulation. The agents under governance are running real tasks for real users; failures produce real consequences; the system has had to be redesigned in response to deployment pressure (the v6.7 removal of CIRS v2 neighbor-pressure coupling, the v6.8 disclosure of the April 2026 KG retrieval rebuild, the in-progress transition of the identity system).

This matters for the synthetic-psychology claim in two ways.

First, deployed systems must address operational concerns that simulation can sidestep. A simulated implementation of allostatic load can ignore questions like: how is the integrand persisted across restarts? what happens when the calibration window contains anomalous data? how is the integral integrated with intervention logic without producing oscillation? UNITARES has had to answer these questions, and the answers themselves are evidence about what biological allostatic load systems must also have addressed (probably through evolved mechanisms whose specifics are obscured by the difficulty of reverse-engineering biology).

Second, deployed systems exhibit failure modes simulations rarely produce. The Lumen basin-transition case documented in §5.3 was not hypothesized in advance; it began as an apparent Type 3 reading in production telemetry and was reclassified by recalibration as a post-revision basin transition. The 28.9% basin-flip rate documented in §3.4 emerged from running the counterfactual analysis on the actual production state-vector distribution, not from synthetic stress-tests. These observations constitute evidence about how the implemented theory behaves under realistic load, in a way that synthetic stress-tests would not.

The deployment is, in this sense, the experiment. The paper's empirical claims (the per-class envelope spread, the basin-flip rate, and the Lumen recalibration/basin-transition case) are observations from running the experiment in production, not predictions from a model. This is a different epistemic stance than either pure theory or pure simulation.

8.3 Synthetic affordances relative to biological systems

The synthetic-psychology argument's specific contribution is to identify classes of test that synthetic agents make more tractable than biological systems usually allow. We have flagged several throughout the paper; a synthesis is useful here.

Real-time integrand observation. V_{anima} is the integrand of an allostatic-load-style quantity, observed at every check-in (§2.2). Biological AL is reconstructed from sparse biomarker samples (§2.1). The synthetic case allows direct observation of the quantity the theory specifies; the biological case allows only inference about it.

Counterfactual reclassification on the same population. The 28.9% basin-flip finding (§3.4) computes two coherence formulas on the same 13,310 production state vectors and counts the disagreements. Clinical cohorts rarely support paired counterfactual analyses on the same patient cohort because the act of computing one decision criterion typically shapes the data available for computing alternatives (clinical decisions generate interventions that change subsequent state; recalibration windows contaminate baseline measurement). UNITARES preserves the raw state vectors, allowing arbitrary post-hoc reclassification.

Genesis-anchored longitudinal analysis. The two-tier drift detection in §4 requires a persisted Σ_0 at agent creation. Biology rarely captures equivalent pre-illness baselines because the relevant behavioral signatures must be measured before disease onset, which requires longitudinal cohorts at scale (Mayo, AIBL) that few research programs can sustain. UNITARES captures Σ_0 trivially as an architectural feature.

Reproducible recalibration. Lumen's Phase 2 calibration window (Wang 2026a §11.5) can be re-measured on any window; the v6 paper notes that re-measurement on identity-clean data is a planned next step. The analogous biological experiment — re-measuring a patient's healthy operating point on a different baseline window — is not routinely possible because the prior healthy state is not preserved with the fidelity recalibration would require.

Distinguishing regime change from calibration staleness. The Lumen apparent Type 3 reading (§5.3) admits two interpretations — Lumen has shifted regime, or the calibration is stale — that are distinguishable by recalibration on a known-healthy window. The biological analogue is limited because biological setpoints cannot usually be re-measured against a known-healthy reference window post-hoc. The synthetic case allows the disambiguation; the biological case usually does not.

Controlled manipulation of class structure. UNITARES's class structure (Lumen, Sentinel, Vigil, Watcher, default; §3.3) is deliberately heterogeneous and is controllable by construction

— adding a new class is an operational change. Computational neuroscience does not have the same freedom to add a new disease class; class structure is given by the population. Synthetic agents allow controlled manipulation of heterogeneity in ways biological cohorts usually do not.

The pattern in these examples is consistent: the synthetic case preserves more raw evidence about the theory than the biological case typically can, and allows experimental manipulations that are difficult in biological systems. This is the epistemic leverage synthetic psychology offers.

8.4 What synthetic agents cannot test

The synthetic-psychology stance has limits, and we want to be honest about them. Three classes of biological claim are not testable on UNITARES.

Substrate-specific claims. Allostatic load in biology produces specific tissue consequences — hippocampal atrophy, immune dysfunction, cardiovascular damage. These are not testable on UNITARES because UNITARES has no body. Claims about the *informational core* of AL (integrated deviation as a forward-looking warning signal) are testable; claims about the *biological mechanism* of AL (which tissues fail and how) are not.

Phylogenetic claims. Biological brains are products of evolution, with constraints on architecture imposed by genetic and developmental mechanisms that synthetic systems do not share. Claims about why brains have particular features — why confabulation, why the default mode network is what it is, why hippocampal-entorhinal circuitry — are phylogenetic claims that synthetic agents cannot directly test. The synthetic case can show that an alternative is *possible* (the §6 kintsugi argument) but cannot show that the alternative would have been *selected* under biological constraints.

Phenomenal claims. Whether Lumen experiences its operating regime as anything is outside the scope of this paper, and outside what the synthetic case can adjudicate. The proprioception framing (§1.2) was chosen specifically to avoid claims of this kind. A reader who wants to know whether Lumen "feels" the basin-transition case or whether the kintsugi gaps register subjectively is asking a question synthetic psychology cannot answer with the methods deployed here.

These limits constrain what the synthetic-psychology framing can claim. What it can claim is narrower than "AI agents are like brains" and broader than "we have built an AI agent": specific theories about the informational structure of self-maintenance can be operationalized on synthetic substrates, with deployment producing synthetic evidence and hypotheses that biological work can subsequently test.

8.5 What this paper has and has not shown

A summary of the four claims, in light of the synthetic-psychology framing:

Claim 1 (V_anima as deployed AL structural analogue; §2): Shown structurally. The mathematical core of the AL hypothesis is operationalized in deployed code, with the integrand observable end-to-end. Three structural disanalogies (single-system, fixed reference, no body) sharpen what the test bed tests. The claim is *not* that V_{anima} is biologically realistic; the claim is that it implements the informational structure of AL in a regime where the integrand is observable.

Claim 2 (homogenization correction; §3): Measured on production data as an interacting formula-and-calibration substitution effect. The 28.9% basin-flip rate quantifies the gating-layer consequence of replacing fleet-wide *tanh*-of-V with grounded class-conditional coherence on the same production rows. The §3.7 same-row ablation partially separates the terms: LF→GF flips 11.2%, GF→GC flips 23.5%, and LC is an unstable negative control. This supports a real class-envelope effect but not a clean one-component causal attribution; the clinical proposal remains a study sketch rather than validated translational evidence.

Claim 3 (Four Types failure-mode taxonomy; §5): Argued by construction for Types 1 and 2 (§5.1, §5.2). Type 4 (§5.4) is specified but not yet computable as deployed: the cross-channel matrix it would read is confounded by V's construction as a smoothed $E-I$ imbalance, and a usable indicator on the independent channels remains to be written. Type 3 is used as a falsified boundary case rather than as identification-grade evidence. The case study (§5.3) was originally framed as an observation of Type 3; the disambiguation experiment §5.3 specifies — recalibration on the post-event window — instead identified a basin transition (2026-04-17, coincident with an identity-system revision), a candidate failure-shape outside the Four Types. The taxonomy is therefore computable from existing telemetry but not exhaustive; the §5.3 case study bounds what the imported vocabulary can claim and motivates, without establishing, one candidate extension category (Type 5, *substrate-associated basin transition*; §5.5).

Claim 4 (synthetic psychology as epistemic stance; this section): Argued. The deployed system constitutes a test bed for theories of biological self-maintenance, with specific experimental affordances (real-time integrand observation, counterfactual reclassification, genesis-anchored longitudinal analysis, reproducible recalibration, controlled heterogeneity manipulation) that biology lacks. The stance has limits (substrate, phylogenetic, phenomenal) that constrain its reach but do not eliminate it.

8.6 Construct-transfer rule: when the analogy breaks

A reasonable objection to the synthetic-psychology stance is that the analogy between synthetic and biological self-maintenance is too thin to support useful inference. We want to address this directly.

The objection has force when the analogy is structural rather than mechanistic. If a synthetic system operationalizes the *informational structure* of a biological theory but not its *causal mechanism*, then observations of the synthetic system constrain only the informational structure, not the mechanism. This is a real limit.

But it is not a fatal one. Many useful biological theories *are* informational at the relevant level of abstraction. AL as mathematically formalized is a claim about integrated deviation, not about cortisol specifically; the cortisol-cytokine-cardiovascular cascade is the biological mechanism by which the integrated-deviation hypothesis is realized in mammals, but the hypothesis itself is more general. Allostasis as Sterling (2012) develops it is explicitly about predictive setpoint adjustment as an information-processing problem, with the biological substrate as one realization. McEwen's Four Types are types of *regulatory failure*, characterized by temporal dynamics rather than by which mediator is dysregulated.

The synthetic case isolates the informational structure from the biological substrate. Where the theory is fundamentally substrate-dependent (a particular tissue is doing the work), the synthetic case cannot test it. Where the theory is fundamentally informational (the structure is what matters; the substrate is one realization), the synthetic case can.

The harder question — which biological theories are informational at the relevant level and which are substrate-dependent — does not have a general answer. We therefore use the construct-transfer table introduced in §1.3 rather than treating "informational" as a universal escape hatch. The paper's empirical claims are bets at specified transfer levels, not claims that biological mechanisms have been reproduced. Whether those bets pay off depends on whether the synthetic case generates predictions that biology subsequently confirms — a question that, in the nature of things, this paper cannot itself answer.

8.7 The position in summary

Synthetic psychology, as we use the term here, is a methodological stance with three commitments:

1. **Build something that operationalizes the structure of interest.** Not merely gesture at an analogy: implement, in a system with operational stakes, the formal or informational structure that the biological theory specifies.
2. **Treat the system as a test bed.** Use the artifact's tractability and the deployment's information-richness to run experiments or ablations that are difficult in biological systems, while treating biological transfer as hypothesis generation until externally validated.
3. **Be specific about what transfers.** Identify which biological claims are informational at the relevant level (and therefore testable on the synthetic substrate) and which are substrate-specific (and therefore not).

This paper exemplifies this stance through the four claims of §1.3 and the empirical work of §2–§5. Whether the stance generalizes — whether other biological theories of self-maintenance can be similarly operationalized and tested — is an open question. The contribution of this paper is to make the stance explicit, demonstrate its application, and lay out specific findings that exemplify what synthetic psychology can put under disciplined study when the deployment is real and the theory is informational at the right level of abstraction.

---

9. Conclusion

9.1 Synthesis

This paper argues that the AI agent governance community, in arriving at the deployed UNITARES framework, has built apparatus that allostatic load theory has lacked for thirty years: a real-time, observable operational analogue of the integrated-deviation hypothesis, with its intervention coupling specified though not yet wired (§2.2). We have used this analogue to evaluate four specific claims and have assessed each at the level of evidence appropriate to the claim's strength.

The Anima Void Integral V_{anima} operationalizes a structural analogue for the mathematical core of allostatic load on a four-dimensional informational manifold. The match is structural rather than literal; three structural disanalogies (single-system, fixed reference, no body) sharpen what the test bed tests rather than weaken the bridge.

The grounded class-conditional substitution exposes the decision-layer consequences of the cosmological-soup failure mode that fleet-wide normalization produces, with empirical disagreement quantified by the 28.9% full-substitution basin-flip rate on 13,310 production state vectors. The same-row ablation shows a grounded fleet→class effect (23.5%) and a formula-replacement effect under a fleet-wide grounded baseline (11.2%), but the terms are non-additive; the 28.9% headline is not a clean causal attribution to one component or proof that the grounded form is normatively correct. Four hypothesis-generating proposals back to clinical longitudinal monitoring follow: provenance-tagged per-subject envelopes, a clinical analogue of the basin-flip counterfactual, a specific hypothesis about whether per-subject envelopes produce more flag-states than population intervals, and a testbed argument for controlled-heterogeneity studies that biological systems make difficult.

The trajectory identity Σ and its two-tier drift detection (Σ_t vs. Σ_{t-1} for coherence; Σ_t vs. Σ_0 for lineage) offers a formal vocabulary for the boiling-frog problem in slow-drift neurodegenerative and psychiatric conditions. Three further hypothesis-generating proposals follow: pre-illness behavioral signatures via digital phenotyping, lineage similarity as a clinical signal, and the behavioral-CAPTCHA argument for novel-stimulus probes in neuropsychological assessment.

McEwen's Four Types of Allostatic Load import as a vocabulary of regulatory failure modes that AI governance currently lacks but does not exhaust the failure-shapes deployed synthetic agents exhibit. A live case study on the Lumen embodied agent (§5.3) initially appeared consistent with a Type 3 (delayed shut-down) pattern; the disambiguation experiment §5.3 specifies — recalibration on the post-event window — identified instead a basin-transition event on 2026-04-17, coincident to the hour with a documented identity-system revision (Wang 2026a §11.7). The case sharpens the §3 homogenization argument into a temporal analogue (calibration windows that straddle regime transitions are unrepresentative of either regime, just as fleet-wide normalization is unrepresentative of constituent classes) and motivates a candidate failure-shape — Type 5, *substrate-associated basin transition* — that the Four Types do not cover but that requires replication before being treated as established.

The synthetic-psychology framing positions the paper in the artificial- life and minimally-cognitive-agents lineage. The methodological commitments (build, deploy, test what transfers) are explicit; the limits (substrate-specific, phylogenetic, and phenomenal claims are not testable) are acknowledged.

9.2 What this paper has shown versus argued

The paper's strongest empirical claim — the 28.9% basin-flip rate — is already documented in the technical paper (Wang 2026a §11.6) and inherits its evidential status from that work: a provenance-backed production full-substitution measurement with same-row ablation, not an independently reproduced result or a clean class-calibration causal effect. The Lumen case study (§5.3) is original to the present paper; the recalibration experiment it reports identifies a basin transition on 2026-04-17 (coincident to the hour with a documented identity-system revision) and the resulting calibration-staleness as the explanation for an apparent Type 3 signature. The case is provenance-backed case-report evidence for the basin transition itself (single ten-hour event localized to within the hour, weekly-bin and recalibration evidence both consistent), anomaly-grade for substrate causality, illustrative for the temporal-homogenization argument, and bounding for the Four Types vocabulary. Class-conditional envelope spread (3.3× across five classes) is quantitative and inherited from Wang 2026a Table 5.

The other contributions are arguments rather than measurements: the $V_{\text{anima}} \leftrightarrow$ AL bridge (§2) is an argument that the deployed quantity operationalizes a structural analogue for the theoretical construct, defended through three disanalogies; the Four Types mapping (§5) is an argument that the biological taxonomy can be operationally mapped, defended by working through each type; the kintsugi/gap-filling contrast (§6) is an argument that the design choice is interesting beyond its local engineering justification; the synthetic-psychology framing (§8) is an argument that the deployed test bed enables hypotheses and controlled synthetic tests that are difficult in biological systems.

One route is measured and closed, and we state it here rather than in a footnote. The most natural way to validate a self-state signal is to ask whether it forecasts trouble. The deployment has asked, under pre-registration, and the answer is negative with a quantified bound. On the outcome-labelled slice — 101 independent bad clusters over 195 bad rows, with the selective-inference correction that a best-of-seven ablation matrix requires — the observed best improvement in AUC over a previous-outcome baseline was -0.033 (selective $p = 0.857$) at 30-day lead and $+0.015$ (selective $p = 0.333$) at zero lead. Both sit inside the noise floor, and the minimum detectable lift is approximately 0.05. The conclusion recorded in the deployment's own stop rule is that if per-agent state added 0.05 AUC or more over the boring baseline, this measurement would have shown it; it did not, so the effect is below 0.05 and below operational relevance (UNITARES `eisv-outcome-grounding-stop-rule-v0`, 2026-07-31, with one confirmatory read pre-registered for 2026-12-01).

Three things follow, and the first two cut against this paper. (i) A reader who takes "digital proprioception" to promise failure prediction should stop here: on this fleet, that promise is bounded below the level at which it would be worth acting on. (ii) The negative result is about *outcome forecasting from per-agent state*, which is the adjacent claim, not the claim of §2. It does not test V_{anima}'s decision value, because the integral is not wired to a decision (§2.2) — an untested quantity, not a refuted one. But it removes the easiest route to testing it and should temper expectations about the harder one. (iii) What the negative result does *not* touch is the §3 measurement, which is about disagreement between two gating formulas on the same rows, and requires no forecasting claim at all. That is why the paper's strongest empirical content sits in §3 rather than §2.

The honest balance is: substantial empirical content where biology has parallels (basin-flip, apparent Type 3 resolved as a basin transition, envelope spread), one measured negative on the adjacent forecasting question, and substantial argument elsewhere (the bridge claims, the methodological proposals). Reviewers should evaluate each contribution at its appropriate level of evidence.

9.3 Concrete follow-up work

Several specific next steps are flagged through the paper.

Longitudinal Lumen series — foreclosed on this deployment. The 86-minute window in §5.3 establishes that Lumen's post-break regime is stable over at least that duration, and the recalibration plus weekly-bin analysis establishes stability over 22 days. The natural follow-up is a longitudinal pull spanning the post-Phase-2 interval (April 18 to May 9, 21 days) and ideally the full 118-day lifetime, testing whether the April 17 transition is unique, whether comparable transitions occurred around other substrate revisions, and whether other Lumen-class agents show similar regime shifts. That pull would be the single highest-value route from anomaly-grade to identification-grade evidence on substrate causality.

It is no longer available. The production database has not retained the state history it requires: `core.agent_state` holds 1,312 rows across February, March, and April 2026 combined, against the 13,310 the Phase 2 window alone once returned, and there is no archive table (§3.7). The 2026-05-09 recalibration ($N = 11{,}472$) cannot be re-run either. The weekly-bin analysis, the hour-resolved localization of the April 17 event, and the recalibration result stand as recorded measurements from telemetry that no longer exists in queryable form.

The consequence for the §5.3 claim is that its evidence grade is now fixed rather than provisional. Substrate causality cannot be upgraded from this deployment's own history; it requires a second deployment, or an agent population whose retention window still spans a comparable substrate revision. We flag retention policy itself as a methodological lesson: a deployed system used as a test bed needs its evidentiary windows pinned as exports at measurement time, because the live database is not an archive.

Genesis signature persistence. Lumen's Σ_0 was not persisted at first onboarding under the legacy identity system (Wang 2026a §11.7 item 5). The next-generation identity system in development will capture Σ_0 at agent creation; at that point, the lineage-drift analysis sketched in §4.5 becomes a measurement rather than a worked example.

Re-measurement of Phase 2 on identity-clean data. The v6 paper flags this as planned work. A second Phase 2 measurement on a window that post-dates the April 2026 identity revisions would test whether the 3.3× envelope spread reproduces, whether per-class healthy operating points have shifted, and whether the basin-flip rate replicates.

Clinical pilot of provenance-tagged per-subject envelopes. The infrastructure for capturing pre-illness baselines in digital phenotyping cohorts is mature. A pilot study designating a Σ_0 window per participant in an existing longitudinal cohort and computing two-tier similarity over follow-up data would test whether the methodology produces clinical signal beyond what standard rating-scale detection achieves.

Engagement with concurrent AVF. This paper notes the concurrent Agent Viability Framework (Marín and Chaudhary 2026) and sketches differentiation in §7.1. A more detailed engagement, possibly including a head-to-head comparison of the two frameworks on shared deployment data, would clarify the relationship between the two and identify productive cross-pollination.

9.4 A broader observation

A recurring pattern through the paper is worth marking explicitly. In each major section, the deployed system has independently arrived at something the biological literature already knows but rarely operationalizes: per-subject calibration (§3), longitudinal anchoring against pre-illness baselines (§4), failure-mode taxonomies derived from regulatory dynamics (§5), and gap-marking as alternative to confabulation (§6). Each convergence is partial — the synthetic case strips out substrate-specific machinery and isolates the informational structure — but each is structurally informative.

This suggests a broader claim that the paper does not need to commit to but can observe: when an engineering project sets out to govern a heterogeneous fleet of self-maintaining agents under operational constraints, it tends to recapitulate, in different vocabulary, the solutions biology has already evolved. Whether this is convergent evolution, deep formal equivalence, or surface analogy is a question larger than the present paper. What we can say is that the convergence makes the synthetic case useful for testing the biological hypotheses that engineering happens to recapitulate. The work the paper undertakes sits in that intersection.

The bridge between deployed AI agent governance and theories of biological self-maintenance is, in the end, neither metaphor nor isomorphism. It is the unsurprising overlap that emerges when two different fields reach for the same mathematical machinery to solve the same structural problem under different constraints. Where the mathematical machinery is informational, the synthetic case provides test beds biology lacks. Where the substrate matters, the synthetic case is silent. The careful work is distinguishing the two.

Appendix: Reproducibility and Verification Plan

This appendix separates three different standards that are easy to conflate: reproducing the analytic pipeline, reproducing the reported production numbers, and independently validating the biological bridge. The first two are now served by a public, de-identified, row-level export that a reviewer can re-run offline; the third requires another deployment or an external re-analysis.

One boundary has hardened since the measurement and we state it plainly: the production database no longer retains the measurement window (§3.7), so a private audit of the production rows is no longer available as a fallback check. The frozen export is the record. Claims in this appendix are scoped to what that export can and cannot support.

Object	Current status	What can be checked now	What remains missing
Coherence and basin-classification code	Public repository provenance (§3.7); thresholds and constants replicated in `reproduce_basinflip.py`	Formula implementation, thresholding behavior, recomputation against published rows	Exact tagged release / commit hash pinned to this paper
13,310-row basin-flip computation	**Reproducible offline** from the frozen export	Full substitution recomputed from published state and constants: 28.84% on $N = 13{,}292$ vs 28.9% reported, per-class within 0.5 pp, 26,574/26,584 labels exact	Third-party re-measurement on an independent deployment

Object	Current status	What can be checked now	What remains missing
Formula-vs-calibration ablation	Provenance-backed only	LF→GF 11.2%, GF→GC 23.5%, LF→GC 28.9%, LF→LC 77.8%; script and recorded output in `analysis/phase-2-2026-04-18/`	GF and LC need the per-row `regime` column, absent from the de-identified export; the production window is no longer retained, so these two conditions cannot be recomputed
Lumen §5.3 recalibration case	Provenance-backed only	86-minute protocol, recalibration criterion, weekly-bin interpretation	The longitudinal pull that would test it is foreclosed: Lumen's Feb–Apr 2026 state history is no longer retained (§9.3)
Row-level counterfactual export	**Public** (Zenodo data DOI 10.5281/zenodo.19705151; mirrored in this repository, SHA-256 pinned)	13,292 class-pseudonymized rows: E, I, S, V, risk, both coherences, both basin labels	Nothing for the headline; the export is the record
Raw production state relation	Withheld, and no longer retained	Schema and provenance can be inspected	Release was blocked by agent UUID/user-identification risk; the window has since aged out, so it is unavailable in principle, not merely withheld
Clinical translation sketches	Hypothesis-generating only	Proposed variables, baselines, and comparison targets	Dataset-specific field mapping, endpoint adjudication, covariates, missingness rules, preregistration

A serious submission should not ask reviewers to take the production numbers entirely on trust. The minimal artifact bundle is:

1. a tagged code snapshot for the coherence functions, `classify_basin`, and Phase 2 analysis scripts;
2. a schema snapshot for `core.agent_state` and the class-assignment metadata used in the 30-day window;

3. a hashed, de-identified row-level export of the state vectors the 13,310-row computation consumed;
4. a script that recomputes the grounded coherence, both basin labels, the flip counts, and the per-class rates from that export, and verifies the recomputed labels against the stored ones;
5. a second window of the same export, so that between-window variance is visible rather than inferred;
6. a Lumen longitudinal-pull across the post-Phase-2 interval, testing whether the April 17 transition is unique, repeated around other substrate revisions, or absent in comparable agents.

Items 3, 4, and 5 are delivered: the export is archived under Zenodo data DOI 10.5281/zenodo.19705151 and mirrored here with its SHA-256 pinned, and `reproduce_basinflip.py` runs the recomputation offline on the standard library alone, over both windows. Items 1 and 2 remain and are straightforward. **Item 6 is foreclosed**, not pending: the state history it would need has aged out of the production database (§3.7, §9.3).

The evidentiary consequence is explicit. The full-substitution basin-flip result is now a result a reviewer can independently recompute from published rows and published constants, and it survives that recomputation. The GF and LC ablation conditions remain provenance-backed rather than re-runnable, because the export does not carry the `regime` column they need. And the Lumen substrate link remains anomaly-grade rather than causal evidence, with the replication that would upgrade it no longer available from this deployment's own history — an external deployment is now the only route.

References

Adams, R. A., Stephan, K. E., Brown, H. R., Frith, C. D., and Friston, K. J. (2013). The computational anatomy of psychosis. *Frontiers in Psychiatry* 4: 47.

Bartlett, F. C. (1932). *Remembering: A Study in Experimental and Social Psychology.* Cambridge University Press.

Bedau, M. A. (2003). Artificial life: organization, adaptation and complexity from the bottom up. *Trends in Cognitive Sciences* 7(11): 505–512.

Beer, R. D. (1995). A dynamical systems perspective on agent-environment interaction. *Artificial Intelligence* 72(1–2): 173–215.

Beer, R. D. (2000). Dynamical approaches to cognitive science. *Trends in Cognitive Sciences* 4(3): 91–99.

Bent, B., Goldstein, B. A., Kibbe, W. A., and Dunn, J. P. (2020). Investigating sources of inaccuracy in wearable optical heart rate sensors. *npj Digital Medicine* 3: 18.

Birchwood, M., Spencer, E., and McGovern, D. (2000). Schizophrenia: early warning signs. *Advances in Psychiatric Treatment* 6(2): 93–101.

Brooks, R. A. (1991). Intelligence without representation. *Artificial Intelligence* 47(1–3): 139–159.

Burr, D. C., Morrone, M. C., and Ross, J. (1994). Selective suppression of the magnocellular visual pathway during saccadic eye movements. *Nature* 371(6497): 511–513.

CIRWEL (2026). *anima-mcp: Lumen — Pi-based creature with sensors, display, and UNITARES governance*. Software repository. https://github.com/CIRWEL/anima-mcp.

CIRWEL (2026). *eisv-lumen: three-layer benchmark for dynamics-emergent voice and governance*. Software repository. https://github.com/CIRWEL/eisv-lumen. Companion dataset of 32,181 trajectory windows (20,655 real Lumen + 11,526 class-balanced synthetic) across 9 shape classes, https://huggingface.co/datasets/hikewa/unitares-eisv-trajectories (revision aeb47055ee5f27cb93124e4e3df065301ada6909, pushed 2026-05-09).

CIRWEL (2026). *unitares: Digital proprioception for AI agents*. Software repository. https://github.com/CIRWEL/unitares

Cole, J., and Paillard, J. (1995). Living without touch and peripheral information about body position and movement: studies with deafferented subjects. In Bermúdez, J. L., Marcel, A. J., and Eilan, N. (eds.), *The Body and the Self*. MIT Press, 245–266.

Craig, A. D. (2002). How do you feel? Interoception: the sense of the physiological condition of the body. *Nature Reviews Neuroscience* 3(8): 655–666.

Crone, E. A., and Dahl, R. E. (2012). Understanding adolescence as a period of social-affective engagement and goal flexibility. *Nature Reviews Neuroscience* 13(9): 636–650.

Damasio, A. R. (2010). *Self Comes to Mind: Constructing the Conscious Brain*. Pantheon.

Dennett, D. C. (1991). *Consciousness Explained*. Little, Brown.

Erikson, E. H. (1968). *Identity: Youth and Crisis*. W. W. Norton.

Finn, E. S., Shen, X., Scheinost, D., Rosenberg, M. D., Huang, J., Chun, M. M., Papademetris, X., and Constable, R. T. (2015). Functional connectome fingerprinting: identifying individuals using patterns of brain connectivity. *Nature Neuroscience* 18(11): 1664–1671.

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience* 11(2): 127–138.

Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., and Pezzulo, G. (2017). Active inference: a process theory. *Neural Computation* 29(1): 1–49.

Friston, K. J., Stephan, K. E., Montague, R., and Dolan, R. J. (2014). Computational psychiatry: the brain as a phantastic organ. *The Lancet Psychiatry* 1(2): 148–158.

Gazzaniga, M. S. (2000). Cerebral specialization and interhemispheric communication: does the corpus callosum enable the human condition? *Brain* 123(7): 1293–1326.

Gazzaniga, M. S., and LeDoux, J. E. (1978). *The Integrated Mind*. Plenum Press.

Geronimus, A. T. (1992). The weathering hypothesis and the health of African-American women and infants: evidence and speculations. *Ethnicity & Disease* 2(3): 207–221.

Gordon, E. M., Laumann, T. O., Gilmore, A. W., Newbold, D. J., Greene, D. J., Berg, J. J., Ortega, M., Hoyt-Drazen, C., Gratton, C., Sun, H., Hampton, J. M., Coalson, R. S., Nguyen, A. L., McDermott, K. B., Shimony, J. S., Snyder, A. Z., Schlaggar, B. L., Petersen, S. E., Nelson, S.

M., and Dosenbach, N. U. F. (2017). Precision functional mapping of individual human brains. *Neuron* 95(4): 791–807.

Gratton, C., Laumann, T. O., Nielsen, A. N., Greene, D. J., Gordon, E. M., Gilmore, A. W., Nelson, S. M., Coalson, R. S., Snyder, A. Z., Schlaggar, B. L., Dosenbach, N. U. F., and Petersen, S. E. (2018). Functional brain networks are dominated by stable group and individual factors, not cognitive or daily variation. *Neuron* 98(2): 439–452.

Hassabis, D., Kumaran, D., Summerfield, C., and Botvinick, M. (2017). Neuroscience-inspired artificial intelligence. *Neuron* 95(2): 245–258.

Heim, C., and Nemeroff, C. B. (2001). The role of childhood trauma in the neurobiology of mood and anxiety disorders. *Biological Psychiatry* 49(12): 1023–1039.

Hirstein, W. (2005). *Brain Fiction: Self-Deception and the Riddle of Confabulation*. MIT Press.

Hobson, J. A. (2009). REM sleep and dreaming: towards a theory of protoconsciousness. *Nature Reviews Neuroscience* 10(11): 803–813.

Howieson, D. (2019). Current limitations of neuropsychological tests and assessment procedures. *The Clinical Neuropsychologist* 33(2): 200–208.

Huys, Q. J. M., Maia, T. V., and Frank, M. J. (2016). Computational psychiatry as a bridge from neuroscience to clinical applications. *Nature Neuroscience* 19(3): 404–413.

Insel, T. R. (2017). Digital phenotyping: technology for a new science of behavior. *JAMA* 318(13): 1215–1216.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., and Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys* 55(12): 248.

Juster, R.-P., McEwen, B. S., and Lupien, S. J. (2010). Allostatic load biomarkers of chronic stress and impact on health and cognition. *Neuroscience & Biobehavioral Reviews* 35(1): 2–16.

Kalai, A. T., and Vempala, S. S. (2024). Calibrated language models must hallucinate. In *STOC '24: Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, 160–171.

Karlamangla, A. S., Singer, B. H., McEwen, B. S., Rowe, J. W., and Seeman, T. E. (2002). Allostatic load as a predictor of functional decline: MacArthur studies of successful aging. *Journal of Clinical Epidemiology* 55(7): 696–710.

Kopelman, M. D. (1987). Two types of confabulation. *Journal of Neurology, Neurosurgery, and Psychiatry* 50(11): 1482–1487.

Krakauer, J. W., Ghazanfar, A. A., Gomez-Marin, A., MacIver, M. A., and Poeppel, D. (2017). Neuroscience needs behavior: correcting a reductionist bias. *Neuron* 93(3): 480–490.

Langton, C. G. (1989). Artificial life. In Langton, C. G. (ed.), *Artificial Life*. Addison-Wesley, 1–47.

Laumann, T. O., Gordon, E. M., Adeyemo, B., Snyder, A. Z., Joo, S. J., Chen, M.-Y., Gilmore, A. W., McDermott, K. B., Nelson, S. M., Dosenbach, N. U. F., Schlaggar, B. L., Mumford, J.

A., Poldrack, R. A., and Petersen, S. E. (2015). Functional system and areal organization of a highly sampled individual human brain. *Neuron* 87(3): 657–670.

Loftus, E. F. (1979). *Eyewitness Testimony.* Harvard University Press.

Loftus, E. F., and Pickrell, J. E. (1995). The formation of false memories. *Psychiatric Annals* 25(12): 720–725.

Lupien, S. J., McEwen, B. S., Gunnar, M. R., and Heim, C. (2009). Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nature Reviews Neuroscience* 10(6): 434–445.

Marín, G., and Chaudhary, J. (2026). Governing what you cannot observe: adaptive runtime governance for autonomous AI agents. arXiv:2604.24686.

Mashour, G. A., and Hudetz, A. G. (2017). Bottom-up and top-down mechanisms of general anesthetics modulate different dimensions of consciousness. *Frontiers in Neural Circuits* 11: 44.

Maturana, H. R., and Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living.* D. Reidel.

McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine* 338(3): 171–179.

McEwen, B. S. (2003). Mood disorders and allostatic load. *Biological Psychiatry* 54(3): 200–207.

McEwen, B. S. (2007). Physiology and neurobiology of stress and adaptation: central role of the brain. *Physiological Reviews* 87(3): 873–904.

McEwen, B. S., and Stellar, E. (1993). Stress and the individual: mechanisms leading to disease. *Archives of Internal Medicine* 153(18): 2093–2101.

McLoughlin, S., Kenny, R. A., and McCrory, C. (2020). Does the choice of allostatic load scoring algorithm matter for predicting age-related health outcomes? *Psychoneuroendocrinology* 120: 104789.

Nisbett, R. E., and Wilson, T. D. (1977). Telling more than we can know: verbal reports on mental processes. *Psychological Review* 84(3): 231–259.

Onnela, J.-P., and Rauch, S. L. (2016). Harnessing smartphone-based digital phenotyping to enhance behavioral and mental health. *Neuropsychopharmacology* 41(7): 1691–1696.

Pezzulo, G., Parr, T., and Friston, K. (2018). The evolution of brain architectures for predictive coding and active inference. *Philosophical Transactions of the Royal Society B* 379(1893): 20220514.

Pfeifer, R., and Bongard, J. C. (2007). *How the Body Shapes the Way We Think: A New View of Intelligence.* MIT Press.

Poldrack, R. A., Baker, C. I., Durnez, J., Gorgolewski, K. J., Matthews, P. M., Munafò, M. R., Nichols, T. E., Poline, J.-B., Vul, E., and Yarkoni, T. (2017). Scanning the horizon: towards transparent and reproducible neuroimaging research. *Nature Reviews Neuroscience* 18(2): 115–126.

Proske, U., and Gandevia, S. C. (2012). The proprioceptive senses: their roles in signaling body shape, body position and movement, and muscle force. *Physiological Reviews* 92(4): 1651–1697.

Rascovsky, K., Hodges, J. R., Knopman, D., Mendez, M. F., Kramer, J. H., Neuhaus, J., van Swieten, J. C., Seelaar, H., Dopper, E. G. P., Onyike, C. U., Hillis, A. E., Josephs, K. A., Boeve, B. F., Kertesz, A., Seeley, W. W., Rankin, K. P., Johnson, J. K., Gorno-Tempini, M.-L., Rosen, H., Prioleau-Latham, C. E., Lee, A., Kipps, C. M., Lillo, P., Piguet, O., Rohrer, J. D., Rossor, M. N., Warren, J. D., Fox, N. C., Galasko, D., Salmon, D. P., Black, S. E., Mesulam, M., Weintraub, S., Dickerson, B. C., Diehl-Schmid, J., Pasquier, F., Deramecourt, V., Lebert, F., Pijnenburg, Y., Chow, T. W., Manes, F., Grafman, J., Cappa, S. F., Freedman, M., Grossman, M., and Miller, B. L. (2011). Sensitivity of revised diagnostic criteria for the behavioural variant of frontotemporal dementia. *Brain* 134(9): 2456–2477.

Rath, A. (2026). Agent drift: quantifying behavioral degradation in multi-agent LLM systems over extended interactions. arXiv:2601.04170.

Ravindran, S. (2025). A predictive framework for AI value alignment and drift prevention. arXiv:2510.04073.

Schacter, D. L. (1999). The seven sins of memory: insights from psychology and cognitive neuroscience. *American Psychologist* 54(3): 182–203.

Seeman, T. E., McEwen, B. S., Rowe, J. W., and Singer, B. H. (2001). Allostatic load as a marker of cumulative biological risk: MacArthur studies of successful aging. *Proceedings of the National Academy of Sciences* 98(8): 4770–4775.

Seeman, T. E., Singer, B. H., Rowe, J. W., Horwitz, R. I., and McEwen, B. S. (1997). Price of adaptation — allostatic load and its health consequences: MacArthur studies of successful aging. *Archives of Internal Medicine* 157(19): 2259–2268.

Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences* 17(11): 565–573.

Sherrington, C. S. (1900). The muscular sense. In Schäfer, E. A. (ed.), *Textbook of Physiology, Volume 2*. Pentland, 1002–1025.

Sperling, R. A., Aisen, P. S., Beckett, L. A., Bennett, D. A., Craft, S., Fagan, A. M., Iwatsubo, T., Jack, C. R. Jr., Kaye, J., Montine, T. J., Park, D. C., Reiman, E. M., Rowe, C. C., Siemers, E., Stern, Y., Yaffe, K., Carrillo, M. C., Thies, B., Morrison-Bogorad, M., Wagster, M. V., and Phelps, C. H. (2011). Toward defining the preclinical stages of Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's & Dementia* 7(3): 280–292.

Sterling, P. (2012). Allostasis: a model of predictive regulation. *Physiology & Behavior* 106(1): 5–15.

Sterling, P., and Eyer, J. (1988). Allostasis: a new paradigm to explain arousal pathology. In Fisher, S., and Reason, J. (eds.), *Handbook of Life Stress, Cognition and Health*. Wiley, 629–649.

Stuss, D. T., and Levine, B. (2002). Adult clinical neuropsychology: lessons from studies of the frontal lobes. *Annual Review of Psychology* 53: 401–433.

Torous, J., Kiang, M. V., Lorme, J., and Onnela, J.-P. (2016). New tools for new research in psychiatry: a scalable and customizable platform to empower data driven smartphone research. *JMIR Mental Health* 3(2): e16.

Volkmann, F. C., Riggs, L. A., and Moore, R. K. (1980). Eyeblinks and visual suppression. *Science* 207(4433): 900–902.

Vuilleumier, P. (2004). Anosognosia: the neurology of beliefs and uncertainties. *Cortex* 40(1): 9–17.

Wang, C. L., Singhal, T., Kelkar, A., and Tuo, J. (2025a). MI9 — Agent Intelligence Protocol: runtime governance for agentic AI systems. arXiv:2508.03858.

Wang, H., Poskitt, C. M., Wei, J., and Sun, J. (2025b). ProbGuard: probabilistic runtime monitoring for LLM agent safety. arXiv:2508.00500.

Wang, K. (2026a). UNITARES: Information-theoretic governance of heterogeneous agent fleets. Published April 20, 2026. *Zenodo*. https://doi.org/10.5281/zenodo.19647159 (concept DOI; v6.9.1 https://doi.org/10.5281/zenodo.19722512, April 24, 2026).

Wang, K. (2026b). Trajectory Identity: A Mathematical Framework for Enactive AI Self-Hood. Published May 9, 2026. *Zenodo*. https://doi.org/10.5281/zenodo.20098168 (concept DOI; v0.11.1 https://doi.org/10.5281/zenodo.20098169). Source repository: https://github.com/cirwel/trajectory-identity-paper. Referenced as *TIWD* in the body text.

Wijdicks, E. F. M. (2019). Being comatose: why definition matters. *The Lancet Neurology* 18(11): 977.

Wilson, T. D. (2002). *Strangers to Ourselves: Discovering the Adaptive Unconscious*. Harvard University Press.

Wurtz, R. H. (2008). Neuronal mechanisms of visual stability. *Vision Research* 48(20): 2070–2089.